FEATURE REFERENCE · FOR TESTERS AND PRODUCT REVIEWERS

GallerySlim — Feature Reference

Every feature of the GallerySlim Android app and the GallerySlim Receiver laptop program: what it does, where to find it, its rules and limits, and what the user sees.

Android app	GallerySlim 1.0.0 (versionCode 2) · Android 12 or later (minSdk 31, target SDK 36)
Laptop program	GallerySlim Receiver 0.1.0 · Windows 10/11, macOS 13 or later, 64-bit Linux
Date	2026-10-24
Basis	The source code as of this date. Where the design documents and the code disagree, this reference describes what the code does; the differences are listed in Appendix A.

Confidential — internal use. This reference describes behaviour read from the code; it has not been run on a device for this edition. Items marked "not verified on a device" are inferred from the code and need a device test.

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1 How GallerySlim keeps photos safe

GallerySlim frees storage in four ways: it **shrinks** photos and videos into smaller copies that are checked to look the same, it **suggests clean-ups**, it **moves** photos to a computer on the same Wi-Fi, and it **keeps the phone slim** with goals, overnight work and nudges. Every one of these is built on the guarantees below. A tester should treat any break of these guarantees as a severe defect.

THE LADDER OF REGRET: EVERY CHANGE IS REVERSIBLE FIRST

1 · Smaller copy
A new, checked copy is added to the gallery. The original is not touched.

2 · Ready to clear
The original waits. The user can Compare, or tap Keep original to throw the copy away.

3 · Android's trash
Only after the user confirms in Android's own dialog. Restorable for 30 days; Undo in the app.

4 · Delete for good
Only on two paths, each with an in-app confirmation and then Android's own dialog.

THE GUARANTEES

Guarantee	What it means in practice
Nothing half-made is ever visible	A smaller copy is written hidden, decoded to prove it is readable, recorded in the app's queue, and only then made visible in the gallery. If any step fails, the hidden copy is removed and the original stays exactly as it was.
Every copy is checked before it is offered	Photos must reach the preset's floor on SSIM, MS-SSIM and (for screen-like images) edge similarity, and then on SSIMULACRA2, measured against the original. Videos must pass a 5-frame spot check and a banding guard. Lossless re-optimisation keeps every pixel identical. A copy that fails is thrown away and the item is reported as skipped. Exception: **Trim** has no quality check (the result is a cut, not a look-alike); it still gets integrity checks.
Originals leave only through Android's trash dialog	Shrinking, overnight runs, trips, Make room and standing rules never trash or delete anything. They only make copies, queue originals in Ready to clear, or pre-select suggestions. The user always confirms in Android's own dialog.
Special photos are never silently degraded	Motion photos are skipped by normal shrinking; their hidden clip is removed only through a separate consent dialog that offers to export the clips first. Ultra HDR photos only ever get a lossless re-optimisation that keeps pixels and the HDR gain map.
Protected by default	Favourites and photos newer than the protect window (default 30 days; 7 or 90 selectable) start unticked; the user can still tick them by hand. Albums set to "Never touch" are excluded from every shrink path (manual, overnight, trip, Make room, light copy, trim). ID documents and photos the app could not read are never suggested for removal anywhere: Clean up lists, Duplicates, Quick review and Chat media (the "ID shield"). Photos not yet checked are shown as held and are never pre-selected.
Only two ways to delete for good	**D1 — "Free this space now":** originals GallerySlim put in the trash, still in the trash, whose smaller copy is verifiably in the gallery and whose identity fingerprint still matches, re-checked one by one just before Android's delete dialog. Records made before this version (no fingerprint) are never eligible. **D2 — Send the trash to computer:** items GallerySlim put in the trash, still in the trash, and verified by the GallerySlim Receiver *in this sending run* (the laptop re-read its saved copy from disk and its SHA-256 and size match the phone's file). Both need an in-app confirmation and then Android's own dialog. Nothing else deletes permanently.

Guarantee	What it means in practice
The original must be the same file	Every clear, undo, keep and delete first re-checks the original's identity fingerprint (size, modified time and the media library's version). If the original changed since it was shrunk, nothing is done: the item moves to "Can't be cleared safely — original changed" in Ready to clear. Each of these operations is written down before Android's dialog opens and finished (or undone) when the app next starts or returns, so a crash never leaves it half-done.
GallerySlim only deletes its own copies	"Keep original" and Undo delete only the smaller copy GallerySlim made, and only after checking that the original is present (Keep original) or back in the gallery (Undo). Cancelling a contact sheet deletes the sheet image GallerySlim had just made.
30-day undo	Everything GallerySlim trashed can be restored while it is in the trash (30 days): from **Recent changes** (Undo shown only while the original is still in the trash), from the **Undo this run** card, or from the phone's gallery trash. Items deleted for good (D1, D2) cannot be undone and show no Undo.
Honest numbers	Space in the trash is not counted as freed. All tools and Recent changes show "freed so far" and "still in the trash" separately, with the date the trash empties. (Your month counts cleared originals in the month they were confirmed, and says so on screen.)
Everything stays on the phone	No account, no ads, no analytics, no uploads. Text reading, face and label detection run on the phone. The network is used only by Send to computer, over the local Wi-Fi, to a computer whose code the user scanned and confirmed. App data is excluded from Google backups and phone-to-phone transfers.
"Sent" is not "safe"	A browser download only proves the bytes left the phone ("fully sent"); such files may be moved to the trash but never deleted for good. Only the Receiver's on-disk re-check makes a file "verified".
Optional safety vault	When the vault is on, an exact copy of each original is saved to a folder the user chose before it is shrunk. If that copy fails, the item is not shrunk.

EVERY ACTION THAT REMOVES SOMETHING

Action (where)	What is removed	How
"Clear *N* originals (*size*) to the trash" (Ready to clear)	Originals of checked copies	Android's trash dialog; 30-day restore
"Move *N* to trash (*size*)" (Clean up lists)	Selected suggestions	Android's trash dialog
"Move *N* to trash" (Quick review)	Items swiped to trash	Android's trash dialog
"Move *N* duplicates to trash" (Duplicates)	Selected extra copies	Android's trash dialog
"Keep best + contact sheet" (Duplicates, bursts)	All shots except the keeper	Android's trash dialog; sheet image saved first
Chat media swipe review	Items swiped to trash	Android's trash dialog, right after the last swipe
"Move *N* sent files to trash" (Send to computer, step 4)	Files sent in this run	In-app dialog, then Android's trash dialog
"Free this space now" / "Free it now" (D1)	Originals already in the trash	In-app dialog, then Android's **delete** dialog; no undo
"Delete *N* checked files for good" (D2)	Receiver-verified trash items	In-app dialog, then Android's **delete** dialog; no undo
"Keep original" (Ready to clear, Compare)	GallerySlim's smaller copy	Direct, only if the original is present, not in the trash and unchanged

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Action (where)	What is removed	How
"Remove the smaller copy" (Ready to clear, changed originals)	GallerySlim's smaller copy	Direct, only while the original is still in the gallery and the copy is the file GallerySlim wrote
Undo (Undo this run, Recent changes)	GallerySlim's smaller copy	After Android's restore dialog, only if the original is back

KNOWN LIMITATION

HEIC copies lose the GPS location tag (stated in Settings). The date stays correct in the gallery.

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2 Reading this reference

2.1 Conventions

- Text in teal between quotes, such as “Get space back”, is shown in the app exactly like that. Highlighted italic parts such as “Shrink *N* items” are filled in at run time: *N* is a count (shown with thousands separators unless noted), *size* or *X* is a size, *name* a file name, *d MMM* a short date such as "24 Oct".
- Paths are written **Home** › **All tools** › **Big videos**. "Home" is the minimal home screen shown after the first scan; "All tools" is the full dashboard behind it.
- Each feature uses the same sub-headings: **Purpose · Where · How it works · Rules & limits · What the user sees · Safety notes**. "Docs vs code" boxes mark places where the design documents say something different; the reference always describes the code. "Tester note" boxes mark behaviour that looks unintended or needs a device check.
- A **snackbar** is the one-line message at the bottom of the app. **Android's trash dialog** and **Android's delete dialog** are the system's own confirmation dialogs; items in the trash can be restored from the phone's gallery for 30 days.
- The **age** of a photo or video is measured from its date taken, or its date added when the date taken is missing.
- Only the Settings sheet is translated into Hindi; everything else is English (see Hindi).

2.2 Sizes and units

Where	Units	Format
All app screens, dialogs and snackbars	Binary: 1 GB = 1,073,741,824 bytes, labelled "GB"/"MB"/"KB"	≥ 1 GB: one decimal ("4.2 GB"); ≥ 1 MB: whole MB ("850 MB"); ≥ 1 KB: whole KB; else "*n* B"
Home-screen widget, low-space notification	Decimal: 1 GB = 1,000,000,000 bytes	"*12.9* GB" (one decimal)
Your month card, its notification and share text	Decimal	≥ 1,000,000,000 bytes: "*1.2* GB"; otherwise whole MB
Storage goal, Make room, trip targets	Binary (5 GB = 5 × 1,073,741,824 bytes)	As app screens
GallerySlim Receiver page and terminal	Decimal (powers of 1000)	One decimal under 10 ("1.2 GB"), else whole
Browser download page (gs1)	Binary (base 1024)	GB with one decimal, MB or KB whole

> **TESTER NOTE**
> The same free space can read differently on the widget (decimal) and on Home (binary), for example 12.9 GB on the widget and 12.0 GB in the app. This is how the code works today.

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3 Getting started

3.1 First launch and permissions

PURPOSE

Get access to photos and videos, then run the first scan. Nothing is changed at this stage.

WHERE

The app opens on the onboarding screen (no app bar) whenever it does not yet have media access. The same screen replaces Home and All tools until access is granted.

HOW IT WORKS

1. The screen shows the title “Your photos, just lighter” and the body “GallerySlim scans your gallery to show how much space you could reclaim — without deleting anything you haven't approved.”
2. A card lists three promises: “Everything stays on your phone. No cloud, no account.” · “Smaller copies are checked before anything is replaced. Originals go to the 30-day trash only after you confirm.” · “You choose what to shrink or clean up, and can compare before and after.”
3. The button “Allow access to photos” opens Android's permission dialog. Caption: “Android will ask you to confirm. You can change this anytime in Settings.”
4. When access is granted (fully or partly), the scan starts at once.

RULES & LIMITS

Case	Behaviour
Android 13 or later	Asks for "Photos and videos" (READ_MEDIA_IMAGES and READ_MEDIA_VIDEO).
Android 12	Asks for storage read access (READ_EXTERNAL_STORAGE). Android 11 and older are not supported.
Android 14 "Select photos" (partial access)	Treated as granted; the scan covers only the selected items. All tools then shows the row “Showing selected items only — allow all for a full estimate”; tapping it opens the app's system settings page. This row is not shown on the minimal Home.
Denied, Android may ask again	Body changes to “GallerySlim can't see your gallery yet.”; the button becomes “Continue”, which asks again.
Denied permanently	The button becomes “Open settings”, which opens this app's page in Android settings.
Notification permission (Android 13+)	Not asked at first launch. Asked just before starting a shrink, light copy, trim or motion-photo strip, running Make room, planning a trip, starting Send to computer, or turning on “Shrink overnight”. The answer does not block the work; without it the work runs silently.
Background jobs	Every app start schedules: the trash-expiry check (daily), free-space sampling (every 12 h), the monthly report check (daily) and the weekly tidy-up nudge (every 7 days, first after 3 days). Each start also records one free-space sample.

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Case	Behaviour
Every start and every return to the app	Finishes any clear, restore or delete whose Android dialog result never reached the app (for example the app was killed while the dialog was open), using what the media library actually shows, and re-checks the media library's version.

WHAT THE USER SEES

The onboarding screen stays calm in every state (no red or error colours). After access, the scanning screen appears.

SAFETY NOTES

The app only reads media at this stage. The camera permission is not requested here; it is asked only when scanning a computer's code (Send to computer).

> **TESTER NOTE**
> The app does not re-check the permission when the user returns from Android settings. After granting access through "Open settings", the onboarding screen probably stays until the app is fully reopened (not verified on a device).

> **DOCS VS CODE**
> The UI spec's onboarding copy ("…without deleting a single photo or video", "This version only looks — it never changes or deletes anything") is outdated; the code shows the text quoted above.

Code: `MainActivity.kt`, `ui/OnboardingScreen.kt`, `ui/DashboardScreen.kt`

3.2 Scanning (first scan and incremental re-scan)

PURPOSE

List every photo and video, detect special formats, and estimate how much space can be reclaimed.

WHERE

- Runs automatically on every cold start of the app (when access exists) and right after access is granted.
- Manually: **Home** › **All tools** › **Rescan gallery** (last row).
- Also after an Undo of a clean-up or move row in Recent changes.
- Not when the app returns from the background, and not on rotation. A second start while a scan runs is ignored.

HOW IT WORKS

1. Reads all images and videos from the phone's media library: name, type, size, dates, favourite flag, album and date modified.
2. For each JPEG it reads the first 128 KiB (and a Samsung trailer of up to 64 KiB) to detect **motion photos** and **Ultra HDR** photos. For each video it reads codec, resolution, frame rate, bitrate and duration.
3. Each item is put in a savings category and given an estimate (table below).

4. Items that already have a smaller copy waiting or confirmed are left out of the totals; items trashed through the app disappear from the totals at once, without a re-scan.

RULES & LIMITS

Category (as shown)	Which items	Estimated saving
"Camera photos"	JPEG, not a motion photo, not Ultra HDR	35 % of size
"Screenshots & images"	PNG	60 % of size
"Videos"	H.264 videos the shrinker expects to make smaller	From the video bitrate plan; videos that are already lean count as 0; a flat 45 % if the video could not be measured
"Hidden motion-photo video"	Motion photos (checked first)	The declared clip size, else 50 % of the file
"HDR photos"	Ultra HDR JPEG	3 % (lossless only)
Already efficient or protected	HEIC, WebP, HEVC or AV1 video, compact H.264; Ultra HDR in a non-JPEG format; GIF, unreadable JPEGs, unknown codecs and other formats	0

- **Motion photo**: XMP `MotionPhoto="1"` or `MicroVideo="1"`, or a Samsung trailer entry of type MotionPhoto_Data. A Samsung trailer without that entry is not a motion photo.
- **Ultra HDR**: the JPEG's first 128 KiB contain an HDR gain-map marker (`hdrgm:Version` or ISO 21496).
- A JPEG that cannot be read is classed as "other format" and is never shrunk (fails closed).
- **Incremental re-scan**: results are cached on the phone. A cached entry is reused only when the file's size and modified time are both unchanged (and the modified time is known). New or changed files are read again; deleted files drop out. If the cache's version does not match, everything is read again.
- The first scan (empty cache) keeps the scanning screen up for at least 1.5 s; later scans are not held. The screen updates at most every 100 ms.

WHAT THE USER SEES

- A 160 dp progress ring: spinning until the total is known, then filling. Centre: the number scanned so far and "of *N*".
- Status line: "Looking through your gallery…" → "Measuring photos…" → "Measuring videos…"
- Once the running estimate is above zero: "Found *size* to free so far".
- No cancel button. The screen stays on during the scan.

SAFETY NOTES

Scanning is read-only.

> **DOCS VS CODE**
> The UI spec lists a fourth status "Checking for hidden video…", category names "JPEG photos / PNG screenshots / H.264 videos" and a "Scanned 2 min ago" caption. The code has three statuses, the category names above, and no "scanned … ago" caption.

Code: `scan/MediaScanner.kt`, `scan/ScanCache.kt`, `scan/SavingsEstimator.kt`, `scan/SpecialFormatDetector.kt`, `ui/ScanningScreen.kt`

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3.3 Home (the minimal home screen)

PURPOSE

One ring, one sentence, one button: how full the phone is and how much can be got back.

WHERE

The start screen once the scan has finished.

HOW IT WORKS

- **Top bar**: "GallerySlim" on the left; a tune icon (accessibility label "Settings") on the right opens the Settings sheet.
- **Ring** (240 dp): the filled part is the used share of the phone's storage (1 – free ÷ total). It animates over 900 ms (jumps straight there when Android animations are off). Centre: the free space as a number and unit (for example "12.3" "GB", or "850" "MB" below 1 GB) and the word "free".
- Under the ring: "of *total*", followed by " · almost full" when the forecast is 0 days, or " · full in about *N* days" when the forecast is 1–365 days. Nothing is added when there is no forecast or it is more than 365 days.
- **Headline**: "You can get back about *X*" when X > 0, otherwise "Your gallery is already in good shape"; both followed by "without losing a single photo."
- **Button** "Get space back" opens the guided flow. Caption: "A few simple steps. You see everything before anything changes."
- **Ready-to-clear card** (only when at least one smaller copy is waiting): "*size* ready to clear" / "Smaller copies of *N* items are done." Tapping it asks for the app lock (if on), then opens Ready to clear.
- **Quick icons**: "Search" → Search; "Undo" → Recent changes; "All tools" → All tools.

RULES & LIMITS

- X ("can get back") = the shrink estimate of all categories (including protected items and motion clips) + the savings of the copies waiting in Ready to clear + the size of confirmed originals still in the trash.
- The ready-to-clear amount is the saving (original – copy), not the size of the originals.
- Below 5 GB free (binary) the ring and number turn amber.
- The forecast needs at least 3 free-space samples spanning at least 2 days within the last 30 days, and free space must be falling (see Widget and forecast).

WHAT THE USER SEES

As above. There is no other content on Home; everything else is under "All tools".

SAFETY NOTES

Home only navigates; nothing here changes files.

> **TESTER NOTE**
> When nothing can be got back the headline reads "Your gallery is already in good shape" followed by "without losing a single photo." — the second line still shows.

Code: `ui/HomeScreen.kt`, `ui/AppNavHost.kt`, `storage/StorageForecast.kt`

3.4 "Get space back" guided flow

PURPOSE

Walk the user through the gentlest steps first, one decision per screen, built from what is on this phone right now.

WHERE

Home › Get space back

HOW IT WORKS

- Top: progress dots (one per step, the current one wider) and an X ("Close") that returns to Home.
- Each page shows "Step *N* of *M*", an icon, an amount in green (only when above zero), a title, a body, the main button, an optional secondary link at the bottom left, and "Skip" at the bottom right.
- The steps are rebuilt from the phone's state. They appear in this order, and only when their condition holds:

Step title	Shown when	Body	Buttons
"Empty what's already done" Amount: size of those originals	At least one original can be deleted for good (D1 rules) and no shrink batch is running	"These originals already have smaller copies and are waiting in the trash. Free the space now instead of in 30 days."	"Free it now" → dialog "Delete these originals for good?" (below). Secondary: "Send to computer first" → starts "Send the trash to computer".
"Clear the originals" Amount: savings waiting	At least one smaller copy is waiting in Ready to clear	"Smaller copies of *N* items are ready. Compare a few if you like, then clear the originals."	"Review and clear" → app lock (if on) → Ready to clear
"Shrink photos and videos" Amount: total estimate	At least one shrinkable item (below)	"Same look, smaller files. Each copy is checked against its original, and originals wait for you."	"Shrink *N* items" → shrinks all of them and opens the progress screen. Secondary: "Try one first" → shrinks only the item with the largest estimated saving (may be a video).
"Clear the junk"	Always	"Old screenshots, chat forwards, blurry shots and duplicates. You swipe to decide, and the trash keeps them 30 days."	"Start swiping" → Quick review. Secondary: "Chat media" → Chat media.
"Keep it tidy by itself"	The five-question setup has not been completed	"Answer five quick questions and GallerySlim keeps an eye on things. It never deletes without you."	"Set up" → the five questions

- The flow shows the first step not yet done. A main button marks its step done before leaving, so returning to the flow shows the next step. "Skip" marks the current step done.
- Exceptions: "Free it now" counts as done only after Android's delete dialog returns OK; "Shrink *N* items" does not count as done if the Google Photos dialog is cancelled.
- The "done" list survives rotation but resets each time the flow is opened again.

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- **End page**: "That's everything for now" / "Anything you started keeps going, and nothing is removed until you confirm it. Come back any time." / button "Done" → Home.

RULES & LIMITS

- **Shrinkable** here means: category Camera photos, Screenshots & images, Videos or HDR photos; an estimated saving above zero; and not protected (not a favourite, and older than the protect window — default 30 days). Motion photos are never part of this step. Items in "Never touch" albums are counted here but dropped when the batch starts.
- "Try one first" picks the single shrinkable item with the largest estimated saving.

WHAT THE USER SEES

Dialog for "Free it now": title "Delete these originals for good?"; body "These *N* originals already have a checked smaller copy in your gallery. Deleting them frees *size* right away. This can't be undone; the smaller copies stay."; buttons "Delete for good" (red) and "Cancel". Confirming opens Android's delete dialog; on OK the snackbar shows "Deleted *N* originals from the trash" (N = what Android actually deleted). If nothing is still eligible at the final re-check, no Android dialog opens and the snackbar shows "Nothing can be deleted for good right now — those originals stay in the trash until it empties".

Dialog before "Shrink *N* items" (only if Google Photos is installed and the warning was never accepted): title "Google Photos backup detected"; body "Smaller copies are new files, so Google Photos will back them up again. Originals you clear stay restorable for 30 days."; buttons "Got it, continue" and "Cancel". Cancel does not record the warning, so it shows again next time.

SAFETY NOTES

The flow never deletes on its own: every removal goes through a dialog. "Free it now" follows the D1 rules (see Free this space now).

> **TESTER NOTE**
> "Try one first" does not show the Google Photos warning, while "Shrink *N* items" does.

Code: `ui/AppNavHost.kt` (GetSpaceBackRoute), `ui/GetSpaceBackFlow.kt`

3.5 All tools (dashboard)

PURPOSE

The full dashboard: every tool, the storage card, the trash ledger and the shrink breakdown.

WHERE

Home › **All tools**. Header: back arrow, title "All tools", a search icon ("Search photos") and a tune icon ("Settings").

HOW IT WORKS

Elements from top to bottom (each only when its condition holds):

Element	Shown when	Content and action
Last-run receipt	A batch was cleared from Ready to clear during this app session	"Cleared *N* originals (*size*) to the trash", "Undo", X ("Dismiss"). See Undo this run.

Element	Shown when	Content and action
Partial-access row	Android 14 partial photo access	"Showing selected items only — allow all for a full estimate" → app's system settings
Storage card "Your phone"	Always	"*size* free" (amber below 5 GB) and a bar; "*used* used of *total*" plus " · almost full" or " · full in about *N* days at this rate" (N ≤ 365). Button "Free up space" → Make room, caption "Shows a plan first, starting with steps that keep every photo."
Trash line in the storage card	At least one confirmed shrink original is still in the trash	"*size* is waiting in the trash" + " and comes back from *d MMM*" when the expiry is known. Links "Free this space now" (only when some originals are eligible and no batch is running) and "Send to computer first".
Ready-to-clear banner	At least one copy waiting	"Ready to clear: *N* originals" / "Their smaller copies are done. Clear the originals to get *size* back." → app lock → Ready to clear
Rules card	Five-question setup not done	"Let GallerySlim tidy for you" / "Answer five quick questions. Nothing is ever deleted without you." → the five questions
Section "Shrink"	Always	Subtitle "Smaller files that look the same. Every photo stays." Then either the empty state (estimate below 100 MB): "Your gallery is already slim" / "Everything is stored efficiently — there's less than 100 MB to reclaim.", or a hero card: "Shrinking can save", a number counting up over 900 ms, "of *size* scanned · *N* photos · *N* videos", a coloured bar and "Each smaller copy is checked against its original. Originals go to the trash only when you say so." Category rows follow (see Category screens).
Section "Clean up"	Always	Subtitle "Remove what you don't need. You choose, and the trash keeps it 30 days." Rows: "Suggestions" ("Old screenshots, chat forwards, blurry shots, duplicates") → Clean up; "Quick review" ("Swipe left to keep, right to clear"); "Chat media" ("WhatsApp and other chat apps"); "Big videos" ("Make lighter copies or trim them").
Section "More"	Always	"Search photos" ("Find a photo by what's in it"); "Recent changes" ("Undo anything from the last 30 days"); "Moved to computer" ("See where moved photos went"); "Your month" ("What you freed, as a card to share"); "Album rules" ("Protect an album, or keep lighter copies of old photos"); "How it works" ("Is it safe? Where do originals go?").
Ledger lines	Lifetime saving > 0 or something in the trash	"Freed so far: *size*"; or, when nothing has left the trash yet and nothing is in it, "You've freed *size* with GallerySlim". Plus "Still in the trash: *N* originals, *size*" + " · frees automatically from *d MMM*" when something is in the trash.

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Element	Shown when	Content and action
"Already efficient or protected"	Always	Total size in grey. Collapsed by default (expanded when the estimate is below 100 MB). Rows with "*N* items": "Already small: modern photo and video formats", "Ultra HDR — kept untouched", "Other formats — left alone"; footer "GallerySlim never touches these without asking."
"Rescan gallery"	Always	Runs the scan again

RULES & LIMITS

- The last-run receipt lives in memory only; it disappears when the app process ends.
- The safety vault has no row here; it is reached from Settings → "Open vault".

WHAT THE USER SEES

For "Free this space now" and "Send to computer first", see Free this space now and Send the trash to computer.

SAFETY NOTES

The dashboard only navigates or opens dialogs; nothing is removed without a dialog.

> **DOCS VS CODE**
> The UI spec's "Favorites & last 30 days — excluded" row and the eyebrow "You can free up to" do not exist; favourites and recent items are counted inside the shrink estimate, and the hero reads "Shrinking can save".

Code: `ui/DashboardScreen.kt`, `ui/AppNavHost.kt`

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4 Shrinking

4.1 Shrinking at a glance

PURPOSE

Make smaller copies of photos and videos that are checked to look the same. Each copy is added to the gallery as a new item; the original is never changed here and waits in Ready to clear.

WHERE

- **Home** › **Get space back** → step "Shrink photos and videos" → "Shrink *N* items" or "Try one first".
- **Home** › **All tools** › **Shrink** section → a category row → category screen → "Shrink *N* items".
- Also: Big videos (light copy, trim), Make room, overnight shrink and trip mode (each described in its own section).

HOW IT WORKS

1. The notification permission is requested if needed, the batch starts and the progress screen opens.
2. Items already waiting in Ready to clear are silently dropped. Items in "Never touch" albums are dropped with a snackbar.
3. Each item gets a preset: the album's rule or cooling (old photos → Light copy), otherwise the global preset from Settings.
4. If the safety vault is on, the original is first copied to the vault; if that fails, the item is not shrunk.
5. The item is routed (table below), encoded, checked and, if it passes, published and queued in Ready to clear.

RULES & LIMITS

Input	What happens
JPEG, PNG	Photo shrink to HEIC (lossless JPEG fallback when HEIC cannot help)
Ultra HDR JPEG	Lossless JPEG re-optimisation only; skipped if it saves less than 1.5 %
Ultra HDR in HEIC, WebP or any other non-JPEG format	Skipped: "Ultra HDR photo — protected"
Motion photo	Skipped: "Motion photo — use the motion-photo flow"
HEIC/HEIF, WebP	Skipped: "Already in an efficient format"
GIF and any other image type; unreadable JPEG	Skipped: "Format not supported"
H.264 video	Video shrink to HEVC (H.265)
HEVC video	Skipped as efficient, except with the Light copy preset when the short edge is above 720 px
AV1 video	Skipped: "Already in an efficient format"
Other or unknown video codec	Skipped: "Format not supported"

- **One batch at a time** across manual shrinks, overnight runs and trips. If the overnight or trip job holds the batch, a manual start shows "Another shrink is already running (*the overnight shrink*). Try again when it finishes." ("the overnight shrink", "the trip shrink" or "another shrink"). A

second manual start while the app's own batch runs is ignored. An overnight run that finds another batch running skips that night; a trip retries after 15 minutes.

- **Low space ordering**: when usable space is below 500 MB or below twice the largest selected item, the batch runs smallest-first; otherwise the user's order is kept.
- **Decision memory**: a photo that earlier gave "Wouldn't get meaningfully smaller" or "Couldn't match the original's quality" with the same size and preset is skipped instantly on a retry, with the same reason.
- Items with a waiting or confirmed copy disappear from the scan lists and totals.

WHAT THE USER SEES

- "Left *N* untouched — their album is set to Never touch" when items were dropped for that reason.
- "Pick a vault folder first" when the vault is on without a folder; the batch is refused (the progress screen still opens and shows "Nothing is running." or the previous result).
- "Something went wrong — see the diagnostic log" if the batch crashes.

SAFETY NOTES

No shrink, strip, trim or light copy trashes or deletes anything. Every copy is written hidden → checked → recorded → made visible.

> **DOCS VS CODE**
> The original design describes Gentle as a JPEG→JPEG (Jpegli) tier and a JPEG XL vault. The code has no Jpegli and no JPEG XL: every preset makes HEIC photo copies (with a lossless JPEG fallback), and the vault stores exact byte copies. Design v2 already reflects the code.

Code: `shrinkflow/ShrinkViewModel.kt`, `shrinkflow/ShrinkController.kt`

4.2 Presets: Gentle, Balanced, Aggressive, Light copy

PURPOSE

One setting decides how hard GallerySlim may shrink: the minimum quality a copy must reach.

WHERE

Home › **Settings** (tune icon) → section "Quality" → "How much to shrink" (radio list).

HOW IT WORKS

The selected preset applies to every manual shrink, overnight run and trip, unless an album rule or cooling gives an old photo the Light copy preset, or Big videos forces Light copy for its videos.

RULES & LIMITS

Preset	Caption shown	Photo SSIM floor	SSIMULACRA2 floor	Start quality	Other effect
"Gentle" (default)	"Highest quality, smaller savings"	0.985	80	87	Video quality target 0.75

Preset	Caption shown	Photo SSIM floor	SSIMULACRA2 floor	Start quality	Other effect
“Balanced”	“Looks the same, good savings (recommended)”	0.97	72	80	Video target 0.65
“Aggressive”	“Biggest savings. Best for screenshots and forwarded photos”	0.93	60	65	Video target 0.5
“Light copy”	“Also reduces size to 2048 pixels. For photos you don't need in full detail”	0.90	55	75	Photos: long edge reduced to 2048 px (only if larger). Videos: short edge reduced to 720 px; HEVC videos above 720p are also accepted. Video target 0.5

Caption under the list: “Shrunk HEIC photos keep their date and rotation, but lose the GPS location tag in this version. JPEG-to-JPEG shrinks keep everything.”

WHAT THE USER SEES

On a fresh install “Gentle” is selected.

SAFETY NOTES

Even Aggressive and Light copy only produce copies that passed their floor; originals still wait in Ready to clear.

> **TESTER NOTE**
> Balanced is captioned "(recommended)" but the default is Gentle.

> **DOCS VS CODE**
> The original design gives SSIMULACRA2 targets of ≥ 90 / 87 / 75. The code uses 80 / 72 / 60 (and 55 for Light copy).

Code: `core/Contracts.kt`, `shrinkflow/ShrinkPrefs.kt`, `res/values/strings.xml`

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4.3 Category screens

PURPOSE

See and choose which items of one kind to shrink (or send to a computer).

WHERE

Home › **All tools** › **Shrink** → a category row. Rows appear only when the category has items and an estimated saving above zero.

Row label	Row subtitle	Screen title
"Camera photos"	"*N* photos → about 35% smaller"	"Camera photos"
"Screenshots & images"	"*N* screenshots → about 60% smaller"	"Screenshots & images"
"Videos"	"*N* videos → about *P*% smaller" (P = estimate ÷ size)	"Videos"
"Hidden motion-photo video"	"*N* photos carry *size* of hidden video"	"Motion photos"
"HDR photos"	"*N* photos → a few % smaller, pixels and HDR untouched"	"HDR photos"

Each row shows the saving as "−*size*".

HOW IT WORKS

- A 3-column thumbnail grid. Videos show a frame, a duration badge (m:ss or h:mm:ss) and the file name. Every tile has a size badge "*size* → ~*size after*" and a round checkbox.
- All non-protected items start selected. Protected items start unticked (see Protected photos).
- Top right: "Select all" / "Clear all" (toggles every non-protected item).
- Bottom bar: "Save about *size*" (hidden when nothing is selected) and the main button "Shrink *N* items" — on Motion photos it is "Remove hidden video from *N* photos". Disabled when nothing is selected.
- Above the bar: "Try on one photo" on the left (see below) and "Send *N* to computer" on the right (when at least one item is selected; queues the selection and opens Send to computer).

RULES & LIMITS

- Estimates per item: JPEG 35 %, PNG 60 %, H.264 from the bitrate plan (45 % if not measured), HDR 3 %, motion clip = declared clip size or 50 % of the file.
- Items in "Never touch" albums are not marked on this screen; they look selectable and are dropped only when the batch starts.

WHAT THE USER SEES

The progress screen after tapping the main button. The Google Photos warning may appear first (see below).

SAFETY NOTES

Selecting and shrinking never removes anything; originals go to Ready to clear.

Code: `ui/CategoryDetailScreen.kt`, `ui/DashboardScreen.kt`

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4.4 Protected photos

PURPOSE

Favourites and recent photos are left alone unless the user deliberately picks them.

WHERE

Window setting: **Settings** › **Safety** › **Protect new photos for** with chips “7 days”, “30 days”, “90 days”; caption “Photos newer than this stay unticked unless you pick them.”

HOW IT WORKS

- An item is protected when it is a favourite, or its age is below the window. Unknown age is not protected.
- On category screens protected items are shown unticked with a chip “Protected · favourite” or “Protected · taken in the last *N* days”, and a header note: “Protected: taken in the last *N* days or a favourite, so left unticked. Tap a photo to include it anyway.”
- The user can tick a protected item by hand. “Select all” never includes protected items.

RULES & LIMITS

- Default window: 30 days. Choices: 7, 30, 90.
- Get space back, Make room, overnight and trips exclude protected items entirely (they cannot be included there).
- Albums set to “Never touch” (see Album rules) are removed from every shrink, light copy and trim at batch start. They are **not** removed from the motion-photo strip.

WHAT THE USER SEES

As above.

SAFETY NOTES

Protection only changes the default selection; it never deletes or hides anything.

Code: `shrinkflow/ExclusionRules.kt`, `ui/CategoryDetailScreen.kt`

4.5 Try on one photo / Try one first

PURPOSE

Let a cautious user shrink just one item and compare it before doing the rest.

WHERE

- Category screens: “Try on one photo” — only on Camera photos, Screenshots & images and HDR photos (never Videos or Motion photos).
- Get space back, step "Shrink photos and videos": “Try one first”.

HOW IT WORKS

- Category screen: the trial item is the selected item with the largest estimated saving; if nothing is selected, the largest non-protected item. The link is hidden when nothing is selected and everything is protected.

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- Get space back: the shrinkable, non-protected item with the largest estimated saving (this may be a video).
- That single item is shrunk; the progress screen then offers "See the difference".

RULES & LIMITS

On category screens the Google Photos warning applies (Cancel aborts the trial). In Get space back the warning is not shown for "Try one first".

WHAT THE USER SEES

Progress screen with one item, then "See the difference" and "Compare the original and the smaller copy before you clear anything."

SAFETY NOTES

Same as any shrink: the original waits in Ready to clear.

Code: `ui/CategoryDetailScreen.kt`, `ui/AppNavHost.kt`

4.6 Google Photos backup warning

PURPOSE

Warn that smaller copies are new files that Google Photos will back up again.

WHERE

Before a shrink or motion-photo strip from a category screen, and before "Shrink *N* items" in Get space back.

HOW IT WORKS

Shown only if the Google Photos app is installed and the warning has never been accepted. It counts as accepted only after "Got it, continue"; Cancel or dismiss means it appears again next time.

RULES & LIMITS

Shown at most until accepted once. Not shown for Get space back's "Try one first", Big videos, Trim, Make room, overnight or trips.

WHAT THE USER SEES

Place	Title / body / buttons
Category screens	"Google Photos backup detected" / "Shrunk files are new files, so Google Photos will back them up again as new items. Originals you later trash stay restorable for 30 days." / "Got it, continue", "Cancel"
Get space back	"Google Photos backup detected" / "Smaller copies are new files, so Google Photos will back them up again. Originals you clear stay restorable for 30 days." / "Got it, continue", "Cancel"

SAFETY NOTES

Informational only.

Code: `shrinkflow/ShrinkViewModel.kt`, `ui/CategoryDetailScreen.kt`, `ui/AppNavHost.kt`

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4.7 Progress screen

PURPOSE

Show a running batch and its result, and explain anything that was not shrunk.

WHERE

Opens automatically after starting any shrink, strip, light copy, trim or Make room run.

HOW IT WORKS

State	What is shown
Running	Header "Shrinking"; a 48 dp ring; "*done* of *total*"; the current file name; "New files are verified before anything is queued. Originals stay untouched."; "About *size* to free once you clear the originals" (when above zero); red "Cancel" with "Finished files stay queued; nothing is deleted." There is no pause.
No batch	"Nothing is running."
Cancelled	Header "Batch finished"; "Cancelled. Files already shrunk are waiting in Ready to clear." Breakdown and buttons as for Finished.
Finished	Header "Batch finished"; "Done. *N* shrunk · *N* skipped · *N* failed".

- **Why some weren't shrunk** card (when anything was skipped or failed): up to 5 reason groups, largest first, each "Skipped × *N* — *reason*" or "Failed × *N* — *reason*" (failures in red), with up to 2 example file names; then the link "Share diagnostic log".
- **Compare links** (when at least one item was shrunk): with exactly 1, "See the difference" and "Compare the original and the smaller copy before you clear anything."; with more, "Compare *name*" for the first 3. These ask for the app lock.
- **Main button**: "Go to Ready to clear" if at least one succeeded, else "Back to dashboard".
- The back arrow returns to Home; the batch keeps running.

RULES & LIMITS

- While a manual batch runs, a foreground service keeps it alive with the ongoing notification "Shrinking your gallery" / "*done* of *total* · *name*" (or "Scanning…" before the first update), channel "Overnight shrink progress". It has no tap action.
- The service uses Android's media-processing type on Android 15 and later (data-sync type before). When Android's daily time limit for background work is reached, the batch stops and a notification says "Android's daily time limit for background work was reached. What finished is in Ready to clear; the rest continues next time."

WHAT THE USER SEES

As above.

SAFETY NOTES

Cancel keeps the finished copies queued; nothing is deleted.

> **TESTER NOTE**
> "Go to Ready to clear" asks for the app lock (if on). Swiping the app away from Recents probably cancels a running manual batch (not verified on a device).

Code: `ui/ShrinkProgressScreen.kt`, `shrinkflow/BatchKeepAliveService.kt`

4.8 Skip and failure reasons

PURPOSE

Tell the user, in plain words, why an item was not shrunk.

WHERE

Progress screen → "Why some weren't shrunk" card.

HOW IT WORKS

"Skipped" means the item was deliberately left as it is. "Failed" means something went wrong and the item was left untouched.

RULES & LIMITS

Kind	Text shown	Typical cause
Skipped	"Ultra HDR photo — protected"	Ultra HDR that lossless re-optimisation could not shrink by 1.5 %, or Ultra HDR in a non-JPEG format
Skipped	"Motion photo — use the motion-photo flow"	Motion photo in a normal shrink
Skipped	"Already in an efficient format"	HEIC, WebP, HEVC, AV1
Skipped	"Format not supported"	GIF, unknown formats, unreadable JPEG, unknown video codec
Skipped	"Couldn't match the original's quality"	No encode reached the preset's floor; video spot check failed
Skipped	"Wouldn't get meaningfully smaller"	Photo saving below 10 %; video saving below 25 % by plan; output not smaller
Skipped	"Not enough free space to make the smaller copy — free some space or shrink smaller files first"	Free-space check before encoding
Failed	"Couldn't protect it in the vault — left untouched"	Vault copy failed
Failed	"Already waiting in Ready to clear"	The item got a copy queued in the meantime
Failed	"The original changed while it was being shrunk — left untouched"	The original was edited, replaced or removed during the work
Failed	"Not enough free space"	Space ran out during work
Failed	"Video conversion stalled — left untouched"	No progress for 120 s
Failed	"Video conversion failed — left untouched"	Empty output, audio lost, duration off by more than 1 s
Failed	"Couldn't convert this photo — left untouched"	Photo encode/publish error
Failed	"Couldn't remove the hidden video — left untouched"	Motion-photo strip error
Failed	"Couldn't trim this video — left untouched"	Trim error
Failed	"Couldn't process this file — left untouched"	Anything else (including lossless-path errors)

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WHAT THE USER SEES

Grouped counts as described on the progress screen.

SAFETY NOTES

Every failure text ends in "left untouched": the original is never modified.

Code: `shrinkflow/ShrinkController.kt`

4.9 Photo quality checks and output

PURPOSE

The testable rules behind a photo copy: when it is accepted, what it looks like on disk, and what metadata it keeps.

WHERE

Applies to every photo shrink (manual, overnight, trip, Make room).

HOW IT WORKS

1. **Too compact for HEIC**: a JPEG with estimated quality ≤ 65 and below 0.8 bits per pixel goes straight to the lossless path.
2. **Free space**: needs at least 2 × the original's size + 300 MB free, otherwise skipped (space reason).
3. **Start quality**: the preset's start quality, lowered by 15 if the source JPEG quality is below 70, or by 8 if below 85; kept within 40–95.
4. **Quality search**: at most 6 encodes, step 8 (16 if the size did not change). A failed check steps quality up; a pass that is not small enough steps it down. It stops early when an encode is at least 1.5 × the original, or when the next step up — predicted as 1.84 × the last encode's size — could no longer save space (would exceed 90 % of the original), so that encode is not attempted.
5. **Check**: the copy is scored at ≤ 512 px on the long edge. Score = the lowest of SSIM, 5-scale MS-SSIM and (for screen-like images, where at least 60 % of 8×8 blocks have 4 or fewer brightness levels) edge similarity; it must reach the preset's SSIM floor. Then SSIMULACRA2 must reach its floor (if it cannot be computed, the SSIM result decides).
6. **Accept** the smallest passing encode that is at most 90 % of the original (saves at least 10 %).
7. **Lossless fallback** (when HEIC fails or is hopeless, and for Ultra HDR): Huffman re-optimisation of baseline, single-scan, 8-bit JPEGs; pixels bit-identical; everything after the image (Ultra HDR gain map, Samsung trailer) copied byte for byte; kept only if it saves at least 1.5 %. Progressive, 12-bit, arithmetic-coded and multi-scan JPEGs are not supported.
8. **Publish**: written hidden, decoded to prove it is readable, recorded, then made visible.

RULES & LIMITS

Property	HEIC copy	Lossless JPEG copy
File name	Original base name + `.heic` (e.g. `IMG_1.jpg` → `IMG_1.heic`)	Same name as the original (the media library de-duplicates it)
Folder	Same folder as the original when it is under DCIM or Pictures; otherwise `Pictures/GallerySlim/`*`album`*`/`	
Date in the gallery	Date taken copied from the original	
Orientation	Baked into the pixels (shows upright)	Kept (EXIF unchanged)

Property	HEIC copy	Lossless JPEG copy
GPS / other EXIF	**Lost** (no EXIF written)	All kept
Quality score in Compare	The measured score	1.0 ("Visually identical · Match 100.00%")

WHAT THE USER SEES

A new item next to the original in the gallery. In Ready to clear, the row shows "*original size* → *new size*".

SAFETY NOTES

A copy that fails any check is discarded; the original is opened read-only.

> **DOCS VS CODE**
> The original design (§6 rule 3) says date, GPS and orientation are copied to every output, and that Ultra HDR is skip-only. In code, HEIC copies carry no EXIF (date kept only in the gallery's date-taken field; GPS lost), and Ultra HDR JPEGs get the lossless re-optimisation.

Code: `shrink/image/JpegToHeicShrinker.kt`, `QualitySearch.kt`, `LumaSsimVerifier.kt`, `JpegOptimizer.kt`, `JpegQuality.kt`, `shrink/PublishPath.kt`

4.10 Video shrinking rules

PURPOSE

Turn H.264 videos into smaller HEVC videos that pass a visual spot check.

WHERE

Category "Videos", Big videos (light copy), Get space back, Make room, overnight and trips.

HOW IT WORKS

1. **Plan a bitrate** from the short side of the picture: ≤ 500 → 1.2 Mbps; ≤ 800 → 2.5 Mbps; ≤ 1200 → 5 Mbps; ≤ 1600 → 9 Mbps; ≤ 2400 → 16 Mbps; larger → 32 Mbps; × 1.5 at 48 fps or more.
2. **Skip** with "Wouldn't get meaningfully smaller" if the planned bitrate is above 75 % of the source bitrate (would save under 25 %); otherwise use the lower of the plan and 55 % of the source. If both resolution and bitrate are unknown, skip.
3. **Content factor**: the target is multiplied by 0.7–1.0 depending on how detailed 3 sample frames (at 25/50/75 %) are.
4. **Screen recordings** (name starts with `screen_recording`, `screenrecord` or `screen-`, or album contains "screen record"/"screenrecord") above 32 fps are re-encoded at 30 fps.
5. **Free space**: estimated output = (target bitrate + 256 kbps) × duration × 1.15; needs 2 × estimate + 300 MB free.
6. **Encode** with the phone's hardware HEVC encoder (constant quality when available, else the planned bitrate). Audio is copied unchanged.
7. **Watchdog**: if no progress for 120 s (checked every 5 s), the encode is cancelled ("stalled").
8. **Checks**: output not empty; audio kept if the source had audio; duration within 1000 ms of the source; output smaller than the source; **spot check** of 5 frames at 10/30/50/70/90 % — every frame must score at least 0.90; **banding guard** — a frame fails if its banding index is at least 1.5 % and more than 1.0 point above the source's.

RULES & LIMITS

- Output name: `base name.mp4` (HEVC in MP4); the media library may de-duplicate it to e.g. "name (1).mp4" (not verified on a device).
- Folder: same as the original when under DCIM, Movies or Pictures; otherwise `Movies/GallerySlim/album/`. Date taken copied.
- A video whose duration is unknown always fails the spot check.
- Leftover work files older than 6 hours are cleaned up.
- The output is always HEVC; AV1 encoding is not used.

WHAT THE USER SEES

Progress per item; reasons as listed in Skip and failure reasons. In Compare, the stored score is the lowest spot-check frame score.

SAFETY NOTES

A failed or stalled conversion leaves the original untouched and no copy visible.

> **DOCS VS CODE**
> The original design mentions VMAF and AV1 where hardware allows. The code uses the SSIM spot check plus banding guard, and never encodes AV1.

Code: `shrink/video/HevcVideoShrinker.kt`, `BitratePlanner.kt`, `VideoQuality.kt`, `Banding.kt`, `SpaceBudget.kt`

4.11 Motion photos: remove the hidden video, export clips

PURPOSE

Motion photos hide a short video inside the photo file. With the user's explicit consent, GallerySlim removes the clip (optionally saving it to a folder first); the still photo stays pixel-for-pixel identical.

WHERE

- **Home** › **All tools** › **Hidden motion-photo video** → screen "Motion photos" → select → "Remove hidden video from *N* photos".
- Export folder: **Settings** › **Other** › **Motion-clip export folder** ("Where the short video inside motion photos is saved before it's removed"; shows the folder name or "Not set").

HOW IT WORKS

1. All non-protected motion photos start selected. "Try on one photo" is not offered here.
2. The Google Photos warning may appear (category-screen wording).
3. Consent dialog: title "Remove hidden video?"; body "Each of these photos carries a short video clip inside the file. Removing it keeps the photo pixel-for-pixel identical but deletes the clip. You can export the clips to a folder first." — plus "No export folder is set yet." when none is set.
4. Buttons: "Export clips, then remove" (when a folder is set) or "Choose export folder" (when not; this only opens the folder picker and closes the dialog — the user must tap the main button again); "Cancel"; and red "Remove without export".
5. The progress screen opens. For each photo: the clip is exported first (if chosen) as `photo base name.mp4`; then a new still is published; the original waits in Ready to clear under "Motion photos".

RULES & LIMITS

- The exported clip must contain a video track; if export fails, the partial clip is deleted and the photo is not stripped.
- The new still is an exact byte prefix of the original: all EXIF/XMP/GPS and any HDR gain map are kept; the motion-photo flag is set to 0; it must decode fully before it becomes visible. Same file name (de-duplicated by the media library) and folder rules as photo copies; date taken copied.
- If no clip boundary can be found, the item fails and the file is untouched.
- Album "Never touch" rules are not applied to the strip. No free-space pre-check was found for this path.

WHAT THE USER SEES

- Success: the originals appear in Ready to clear under "Motion photos"; Compare works (score "Not measured").
- Failure: “Couldn't remove the hidden video — left untouched” (or “Not enough free space”).
- Bad export folder: “Couldn't use that folder for clip exports”.

SAFETY NOTES

Normal shrinking never touches motion photos. Removing the clip always needs this consent dialog. The original keeps its clip until it is cleared from Ready to clear, and then stays restorable in the trash for 30 days.

Code: `ui/CategoryDetailScreen.kt`, `shrink/image/MotionPhotoStripperImpl.kt`, `MotionPhotoSplitter.kt`

4.12 Big videos and Light copy (720p)

PURPOSE

List videos biggest first and make 720p "light copies" or trimmed copies.

WHERE

Home › **All tools** › **Big videos** (in the Clean up section). Screen title “Big videos”.

HOW IT WORKS

- Intro: “Your videos, biggest first. A light copy keeps the video at 720p; Trim keeps just the part you choose. Both make a new file. The original stays until you clear it in Ready to clear.”
- Every video is listed, largest first (no minimum size). Each row: thumbnail, name, “*size* · *short edge*p · *m:ss*” (unknown parts omitted), then “720p copy saves about *size*” (green) or “Already light”.
- Nothing is pre-selected. Only rows with an estimated saving can be selected. Header: “Select all” / “Clear all” (only when at least one row can be selected).
- Bottom: “*N* videos · *size* in total” when nothing is selected, else “*N* selected”; button “Make 720p light copies”, which becomes “Make 720p light copies (save about *size*)” once something is selected (disabled when nothing is).
- Per row: “Trim” for H.264/HEVC videos longer than 2 seconds (see Trim).
- The button runs the Light copy preset on the selected videos, whatever the global preset, and opens the progress screen. No Google Photos warning and no confirmation on this path.

RULES & LIMITS

- Light copy: short edge becomes 720 px (aspect ratio kept, even dimensions, rotation respected); smaller videos are not upscaled (an H.264 one is just re-encoded to HEVC).
- No saving (row not selectable) when the codec is not H.264/HEVC, an HEVC video is already 720p or smaller, the planner sees no saving, or size/duration is unknown.
- All video checks (space, spot check, banding, integrity, watchdog) still apply. "Never touch" albums are honoured; album Light-copy rules and cooling are not needed here.

WHAT THE USER SEES

Empty state: "No videos found". Dropped items: "Left *N* untouched — their album is set to Never touch".

SAFETY NOTES

Originals wait in Ready to clear under "Videos".

> **DOCS VS CODE**
> The round-2 plan names the action "Light copy (720p)"; the button reads "Make 720p light copies".

Code: `ui/BigVideosScreen.kt`, `shrink/video/LightCopy.kt`

4.13 Trim

PURPOSE

Keep only part of a video as a new, smaller file.

WHERE

All tools › **Big videos** → "Trim" on a row. Screen title "Trim".

HOW IT WORKS

- Shows the file name and a still frame (not playback) that refreshes when a handle is released.
- A two-handle range slider starts at the full length; values snap to 100 ms (the far end keeps the exact end). Labels "Start *m:ss*" and "End *m:ss*", and "Keeps *m:ss* of *m:ss*".
- Body: "Trim makes a new video with only this part. The full original stays in your gallery until you clear it in Ready to clear, and then sits in the trash for 30 days. After that, the parts you cut are gone."
- Bottom: "Save about *size*" (when the range is valid and saves something) and "Make trimmed copy" (enabled only for a valid range).
- After tapping, the progress screen opens; back returns to Big videos.

RULES & LIMITS

Rule	Message shown
Video length unknown	"This video's length is unknown, so it can't be trimmed."
Start before 0	"Start is before the beginning"
End more than 500 ms past the end (within 500 ms it snaps to the end)	"End is after the video ends"

Rule	Message shown
Start at or after end	"Start must be before end"
Kept part under 1 second	"Keep at least 1 second"
Less than 1 second cut in total (includes the initial full range)	"Move a handle to cut something"

- H.264/HEVC only. Frame-accurate cut, re-encoded to HEVC (planner bitrate for H.264; source bitrate otherwise).
- **No quality spot check**; integrity checks remain (audio kept, duration within 1 s of the chosen length, output smaller than the source).
- Output name: `base_trim.mp4`, same folder and date rules as other video copies.

WHAT THE USER SEES

Refusals: "This video's album is set to Never touch", "This video is already waiting in Ready to clear", "Pick a vault folder first". A trim started while a batch runs is silently ignored. The original appears in Ready to clear under "Trimmed videos"; Recent changes shows "Trimmed".

SAFETY NOTES

The cut parts are lost only after the original is cleared and its 30 days in the trash end.

> **DOCS VS CODE**
> The round-2 plan says trimmed videos have no Compare. In code the "Compare" button still shows on trimmed-video rows; tapping it opens "Can't compare this item".

Code: `ui/TrimScreen.kt`, `shrink/video/VideoTrim.kt`, `HevcVideoShrinker.kt`

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5 Checking and clearing

5.1 Compare: entry points and unavailable states

PURPOSE

Before clearing anything, the user can check an original against GallerySlim's smaller copy by eye. Compare never trashes or deletes; its only action with an effect is "Keep original".

WHERE

- **Ready to clear**: "Compare" on every row (next to "Keep original"). At font scale 1.3 and above both actions move to a second line. While Android's trash dialog is open, taps are ignored.
- **Progress screen** after a batch with at least one success: "See the difference" (exactly 1 shrunk) or up to 3 links "Compare *name*". These ask for the app lock.

HOW IT WORKS

The screen loads the queued record and opens the photo comparer (normal photo shrinks, lossless JPEGs, light copies, stripped motion photos) or the video comparer (video shrinks and 720p light copies). Header "Compare" with a back arrow and a spinner while loading.

RULES & LIMITS

Case	Title	Message (button "Go back")
Record not found	"Item not found"	"This item is no longer in the queue. Nothing was changed."
Already cleared or kept	"Already handled"	"This item is no longer waiting for confirmation, so there is nothing to compare."
Trimmed video (unsupported kind)	"Can't compare this item"	"Comparison isn't available for this kind of item. Nothing was changed."
Both files missing	"Both files are missing"	"Neither file can be found any more. Nothing was changed."
Original missing	"Original is missing"	"The original can't be found — it may already be in the trash. Nothing was changed."
Smaller copy missing	"Shrunk copy is missing"	"The smaller copy can't be found. It will be dropped from the queue before anything is trashed, so your original stays safe."

WHAT THE USER SEES

Back arrow, system back, "Looks good" and "Go back" all return to the screen Compare was opened from.

SAFETY NOTES

"Looks good" approves nothing; originals are cleared only from Ready to clear.

Code: `ui/compare/CompareScreen.kt`

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5.2 Compare: photos

PURPOSE

Judge whether the smaller copy looks the same as the original photo, down to single pixels.

WHERE

Compare on a photo record (see entry points above). Header: back arrow, "Compare", file name.

HOW IT WORKS

- Two modes in a toggle: "Wipe" (default) and "Side by side". The canvas is near-black in both light and dark theme.
- **Wipe**: one image with a vertical divider starting at 50 %. Original on the left (right in right-to-left languages), copy on the other side. Drag the round white handle (44 dp, 48 dp touch strip); it stops 26 dp from each edge. Tapping the canvas does not move it. Corner labels "Original" and "GallerySlim", each fading out when its side is too narrow.
- **Side by side**: two panes with a 2 dp gap, as a row or a column (whichever shows the photo larger; a row on a tie). Original first.
- **Zoom and pan** are shared by both images: pinch to zoom, drag to pan, fling above 400 px/s (no fling when reduce-motion is on).
- **Double-tap**: at fit, zooms toward the tap to actual pixels (kept between 2× and 4× of fit); when zoomed, returns to centred fit.
- **Pill button** bottom-right: "100%" jumps to one photo pixel per screen pixel; at 100 % it reads "Fit".
- **Zoom chip** bottom-left: "*n*%", shown only when zoomed more than 1 % above fit.
- **Touch and hold** (250 ms without moving): shows the original everywhere, with a haptic tick and a top chip "Original"; release to return. A hint chip "Touch and hold to see the original" shows for 3.5 s or until the first touch.

RULES & LIMITS

- Zoom range: from fit (or 1:1 if the photo is smaller than the screen) to the larger of 3× fit and 4× actual pixels.
- Landscape: side panel when the window is wider than tall and at least 560 dp wide (panel 40 % of width, 280–360 dp). Portrait: info panel at most half the height; canvas at least 240 dp.
- Mode, divider position and zoom survive rotation.

Stored quality score	Verdict	Second part
≥ 0.985	"Visually identical"	"Match *XX.XX*%" (truncated, not rounded)
0.97 – 0.985	"Nearly identical"	Match %
below 0.97	"Some loss — compare closely" (warning colour)	Match %
none	"Not measured"	—

WHAT THE USER SEES

- Info card line 1: "*orig* → *new* · *change*", where change is "*N*% smaller" (green), "same size" or "*N*% larger"; a real saving shows at least 1 % and never 100 %. Sizes use binary units with one

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decimal while the number is below 100 (e.g. "4.2 MB", "245 MB", "1.3 GB"); KB are whole numbers.
- Line 2: the verdict (table). Lossless JPEG copies show "Visually identical · Match 100.00%"; stripped motion photos show "Not measured".
- Bottom of the card: columns "Original" and "GallerySlim" with resolution "W × H" and "MP · format" (e.g. "12 MP · JPEG"); unknown values show "—".
- Actions: "Keep original" (red text) and "Looks good" (goes back only). If the pair failed to load, only Keep original remains, and the canvas shows "Couldn't open one of the two photos. It may have been moved or deleted."
- TalkBack: the divider is a slider "Comparison divider" ("Original 45%, GallerySlim 55%"); custom actions zoom to 100 %/fit and show the original only.

SAFETY NOTES

See Keep original.

> **DOCS VS CODE**
> The copy is labelled "GallerySlim", not "Smaller copy". There is no difference/heat-map view and no flicker toggle; only Wipe, Side by side and touch-and-hold.

Code: `ui/compare/PhotoCompareScreen.kt`, `PhotoComparer.kt`, `CompareMath.kt`

5.3 Compare: videos

PURPOSE

Play the original and the smaller copy in sync and judge them by eye.

WHERE

Compare on a video record (normal video shrink or 720p light copy).

HOW IT WORKS

- Top: "Compare", the summary "*orig* → *new* · *change*" and the file name.
- Three-way toggle: "Side by side" (default) | "Original" | "GallerySlim".
- **Side by side**: two panes (row or column), each with a chip "Original" or "GallerySlim" and a speaker icon. One transport drives both. Only one side is audible: the GallerySlim copy by default; tap a pane to hear that side.
- **Original / GallerySlim**: one surface shows only that file; tap the video to switch files. Position and play/pause are kept.
- **Transport**: "Play both" / "Pause" / "Play again" (at the end; restarts both from 0:00); elapsed time, seek bar and total (the shorter of the two durations, or "—"); mute button "Mute" / "Turn sound on".
- "Compare this frame": pauses and reads both files at the same moment, then opens the photo comparer on the two frames (title "Compare frame", subtitle "*name* · *m:ss*", no Looks good / Keep original). While reading the button shows "Reading frame…". Back returns to the video at the paused position.
- Bottom bar: "Keep original" (red outline) and "Looks good". Both pause playback first.

RULES & LIMITS

- The silent pane is re-synced when it drifts more than 100 ms from the audible one, at most every 500 ms.
- "End" means within 50 ms of the end. Seek previews at most every 80 ms; releasing lands on the exact frame and resumes if it was playing.
- Playback pauses when the app loses the foreground or audio focus (e.g. a call); players are rebuilt at the same position on return. The screen stays on while playing.
- Frame grabs: taken at least 50 ms before the end; a frame may use up to 1/8 of app memory (4–48 MB); very large frames are scaled down.
- Verdict uses the same thresholds as photos; the score is the lowest spot-check frame score from shrinking.

WHAT THE USER SEES

- Verdict notes: “You shouldn't be able to tell them apart.” · “Differences are hard to spot at normal viewing size.” · “Pause on a detailed frame and compare closely before you decide.” · “Video quality isn't scored on the phone. Play both and compare by eye.” (when there is no score), followed by "Match *XX.XX*%" when a score exists.
- Table with columns Original and GallerySlim: size, resolution, format (H.264 / HEVC / AV1 / Other / —), bitrate (e.g. "5.0 Mbps"), duration.
- Footer: ““Keep original” discards the smaller copy and leaves your video exactly as it was.”
- Pane errors: “Can't open the original” / “Can't open the GallerySlim copy” + “The file may have been moved or deleted.”; “Can't play the original here” / “Can't play the GallerySlim copy here” + “This phone's player stopped with an error.”
- Frame-read errors (shown 4 s): “Couldn't read this frame from the original.” / “Couldn't read this frame from the GallerySlim copy.”

SAFETY NOTES

Compare plays files read-only.

Code: `ui/compare/VideoCompareScreen.kt`, `VideoCompareController.kt`, `VideoCompareLogic.kt`, `VideoSyncCoordinator.kt`

5.4 Ready to clear

PURPOSE

Every checked copy leaves its original waiting here. The user checks them, keeps any original they want, then moves the others to Android's 30-day trash with one system dialog.

WHERE

- Home → Ready-to-clear card (app lock).
- All tools → banner “Ready to clear: *N* originals” (app lock).
- Get space back → “Clear the originals” → “Review and clear” (app lock).
- Progress screen → “Go to Ready to clear”, and taps on overnight / storage-goal / trip notifications (app lock too).

HOW IT WORKS

1. Header "Ready to clear" with back (always returns to Home) and, when the list is not empty, the link "Send to laptop".
2. Intro: "Every file below has a verified, smaller copy in your gallery. Clearing moves the originals to the trash, where they stay restorable for 30 days."
3. Groups in fixed order: "Photos", "Videos", "Trimmed videos", "Motion photos"; each header shows "*N* · –*size*" (count · saving).
4. Each row: file name, "*original size* → *new size*", "Compare" and "Keep original" (red).
5. Bottom: "Saves *size* once the trash empties (about 30 days) · restorable until then" and the button "Clear *N* originals (*size*) to the trash" (count and total size of the originals).
6. Tapping Clear first re-checks every item. The copy must be a visible gallery item (not pending, not trashed, size above zero, same size as written); the original must be present and still the same file (same size and modified time, and the media library's version unchanged). Copy missing → dropped from the batch and the list, original untouched. Original changed → the item moves to the "Can't be cleared safely" section. Original already gone or in the trash → it leaves the list with its smaller copy kept. If the check itself fails, nothing changes.
7. Android's trash dialog opens once for all remaining originals. OK: the records are confirmed, the batch becomes the "last run" for Undo, and the list empties (no snackbar). Cancel: nothing changes.
8. "Send to laptop" queues the originals of **all** listed records (with their real dates, so they sort into month folders), starts sharing and opens Send to computer. The records stay in Ready to clear.

RULES & LIMITS

- Saving per record = original – copy (never below 0).
- Items waiting here are never shrunk again.
- **Can't be cleared safely — original changed**: a separate section at the bottom lists items whose original was edited, replaced or moved since it was shrunk (found when clearing, keeping, or by the start-up check). Caption: "These originals were edited, replaced or moved since they were shrunk, so GallerySlim won't touch them. You can remove the smaller copy (the original stays) or dismiss." Each row: name, "Smaller copy: *size*", "Dismiss" (hides the row; nothing is deleted) and red "Remove the smaller copy" (deletes only GallerySlim's copy, and only while the original is in the gallery and not in the trash and the copy is the file GallerySlim wrote). These items are never part of the Clear button.
- While the trash dialog is open, Compare, Keep original and Send to laptop are ignored, and the Clear button is disabled.

WHAT THE USER SEES

- Empty: "Nothing to clear yet. Shrink some items and their originals will show up here."
- Some copies missing at the re-check: "Skipped *N* item(s) whose shrunk copy is missing — originals left untouched"
- Originals changed: "*N* item(s) changed since they were shrunk — they can't be cleared safely"; originals already gone: "*N* original(s) were already gone from your gallery — kept their smaller copies"; check failed: "Couldn't check *N* item(s) — try again".
- Remove the smaller copy: "Removed the smaller copy of *name*", "Couldn't remove the smaller copy of *name*", or (original not in the gallery) "Kept the smaller copy of *name* — the original isn't

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in your gallery, so it may be the only version left".

- App lock prompt: "Unlock GallerySlim" / "Confirm it's you to see what's ready to clear and the vault".

SAFETY NOTES

No original is trashed without a published copy; the re-check happens immediately before the dialog.

> **TESTER NOTE**
> The link reads "Send to laptop" here but "Send to computer" elsewhere. Records whose originals were moved to the trash through Send to computer stay in the list (not verified on a device); Keep original then reports that the original is already in the trash.

Code: `ui/PendingConfirmScreen.kt`, `shrinkflow/PendingQueue.kt`, `shrinkflow/ShrinkViewModel.kt`

5.5 Keep original

PURPOSE

Reject a smaller copy: throw it away and keep the original exactly as it was.

WHERE

Ready to clear rows; the photo comparer; the video comparer's bottom bar.

HOW IT WORKS

No confirmation dialog. In Compare the tap acts only while the screen is fully on top (taps during a transition and double taps are ignored), then returns to the previous screen. The app first checks the original:

Original's state	Result	User sees
Present, not in the trash, unchanged	The smaller copy is deleted; the item leaves Ready to clear	The row disappears; no message
Present, but changed since it was shrunk (or the copy changed)	Nothing is deleted; the item moves to "Can't be cleared safely"	"*name* changed since it was shrunk — nothing was deleted"
Present, but deleting the copy fails	Nothing changes	"Couldn't remove the shrunk copy of *name*"
In the trash, gone, or unknown	The copy is kept; the item leaves the queue as confirmed	"*name*: the original is already in the trash — kept the shrunk copy"
The check itself failed	Nothing changes	"Couldn't check the original of *name* — try again"

RULES & LIMITS

Only GallerySlim's own copy is ever deleted, and only when the original is verifiably present.

WHAT THE USER SEES

As in the table.

SAFETY NOTES

This never deletes the only remaining copy of a photo.

Code: `shrinkflow/ShrinkViewModel.kt` (keepOriginal), `ui/AppNavHost.kt`

5.6 Trash ledger: freed vs still in the trash

PURPOSE

Honest numbers: clearing originals frees nothing until Android's trash empties.

WHERE

- Home: the "can get back" figure includes originals still in the trash.
- All tools → "Your phone" card: "*size* is waiting in the trash" (+ " and comes back from *d MMM*").
- All tools, near the bottom: "Freed so far: *size*" / "You've freed *size* with GallerySlim" / "Still in the trash: *N* originals, *size*" + " · frees automatically from *d MMM*".
- Recent changes and Your month (see those sections).

HOW IT WORKS

For each confirmed shrink, the app looks up the original:

State of the original	Counts as
In the trash	"In the trash": the full original size; the earliest expiry date is shown
No longer exists	"Freed": original – copy
Present and not trashed (restored by the user)	Nothing
Lookup error	In the trash (never claims freed space)

RULES & LIMITS

- Only shrink originals are counted. Items trashed by Clean up, Quick review, Duplicates or Chat media are not part of the ledger.
- Refreshed when lifetime savings or the last run change, when Recent changes opens, and after Free this space now.
- "You've freed … with GallerySlim" (lifetime saving) also includes originals the user later restored.

WHAT THE USER SEES

As above.

SAFETY NOTES

Read-only bookkeeping.

Code: `shrinkflow/TrashLedger.kt`, `shrinkflow/ShrinkViewModel.kt`

5.7 Undo this run

PURPOSE

One-tap undo of the batch just cleared from Ready to clear.

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WHERE

Home › **All tools**, first card: "Cleared *N* originals (*size*) to the trash", "Undo", X ("Dismiss"). Not shown on the minimal Home.

HOW IT WORKS

1. "Undo" opens Android's restore-from-trash dialog for the run's originals. Cancel changes nothing and the card stays.
2. After OK, for each record the app checks that the original is back (exists and is not trashed). Only then does it delete GallerySlim's copy and mark the item undone. Otherwise the copy is kept.

RULES & LIMITS

- A "run" is exactly the set confirmed in the last Ready-to-clear trash dialog.
- It lives in memory only: lost when the app process ends; replaced by the next cleared batch; cleared by X, by a completed Undo, and by any single shrink Undo in Recent changes. The 30-day per-item undo is in Recent changes.

WHAT THE USER SEES

"Restored *N* originals" when every copy could be removed, otherwise "Restored *N*; *M* keep their smaller copy because the original isn't verifiably back". Before the dialog, each original must still be in the trash and unchanged; if none is: "Nothing to restore — those originals have left the trash". If the check fails: "Couldn't check some originals — try again".

SAFETY NOTES

A copy is never deleted unless its original is verifiably back.

Code: `ui/DashboardScreen.kt`, `shrinkflow/ShrinkViewModel.kt`

5.8 Free this space now (delete for good, path D1)

PURPOSE

Permanently delete originals GallerySlim already put in the trash, instead of waiting about 30 days.

WHERE

- **All tools** › **Your phone** card → "Free this space now". Shown when originals are in the trash, at least one is eligible, and no shrink/strip batch is running.
- Get space back → "Empty what's already done" → "Free it now" (same conditions).

HOW IT WORKS

1. An in-app dialog (wording below), then Android's own delete dialog.
2. Just before Android's dialog, each record is re-checked. Eligible only if: it is a confirmed shrink; its original is verifiably in the trash (the lookup succeeded); and its smaller copy is verifiably published (visible, not trashed, size above zero).
3. The original must also still be the same file: a recorded identity fingerprint (size and modified time) and the media library version recorded when it was shrunk must both match. Records made before this version have no fingerprint and are never eligible.
4. Anything that fails stays in the trash and expires normally. If nothing is still eligible, no Android dialog opens and the snackbar shows "Nothing can be deleted for good right now — those

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originals stay in the trash until it empties".

RULES & LIMITS

	All tools	Get space back
Title	"Delete these originals for good?"	"Delete these originals for good?"
Body	"These *N* originals already have a verified smaller copy in your gallery. Deleting them from the trash frees *size* right away. This can't be undone — the smaller copies stay."	"These *N* originals already have a checked smaller copy in your gallery. Deleting them frees *size* right away. This can't be undone; the smaller copies stay."
Confirm (red)	"Delete *N* originals"	"Delete for good"
Cancel	"Cancel"	"Cancel"

WHAT THE USER SEES

On OK: "Deleted *N* originals from the trash", then the ledger refreshes. Cancelling either dialog changes nothing.

SAFETY NOTES

No undo. The smaller copies stay in the gallery.

> **TESTER NOTE**
> The success count is the number Android actually deleted. The two dialog wordings differ ("verified" vs "checked"). The operation is recorded before Android's dialog opens; if the app is killed while the dialog is up, it is resolved on the next start.

> **DOCS VS CODE**
> The original design (§6 rule 1) and a code comment say permanent delete exists in "exactly one place". There are two (this one and Send the trash to computer), as the 2026-10-22 addendum and Design v2 (D1, D2) describe.

Code: `shrinkflow/ShrinkViewModel.kt` (freeSpaceIntentSender), `ui/DashboardScreen.kt`, `ui/AppNavHost.kt`

5.9 Recent changes and Undo

PURPOSE

Everything GallerySlim changed in the last 30 days, with Undo on each row.

WHERE

Home › **Undo**, or **All tools** › **More** › **Recent changes**. No app lock.

HOW IT WORKS

- Header: "Freed so far: *size*" (always shown); if anything is in the trash, "Still in the trash: *size*. That space comes back when the trash empties, and until then everything below can be undone."
- Rows newest first, grouped under date headers. Each row: name and "*detail* · *size*", and "Undo".

Row type	Source	Detail text	Size
Shrink	Confirmed shrinks whose copy was made in the last 30 days, plus older ones whose original is still in the trash; also shrinks whose original was deleted for good by Free this space now	"Trimmed" for trims; "Shrunk, original cleared" while the original is in the trash; "Shrunk, original gone from the trash" once it has left; "Shrunk, original deleted for good" after Free this space now	Saving
Logged	Clean-ups, duplicates, contact sheets, chat media and moves in the last 30 days, not undone	"Moved to *computer*", "Moved to *computer*, deleted from the phone" (after delete for good), "Folded into a contact sheet", or "Cleared · *reason*"	File size

- Reasons in "Cleared · …": the Clean up list title; "Duplicate or burst extra"; the Quick review reason; the chat folder name (or "Chat media"); "Burst" for contact sheets.
- **Undo on a shrink row**: shown only while the original is still in Android's trash. Android's restore dialog for that original, then the same safe copy removal and messages as Undo this run (also clears the Undo-this-run card).
- **Undo on a logged row**: shown unless the item was deleted for good. Before the dialog the app checks the item is still in the trash (otherwise "Nothing to restore — those files have left the trash"). On OK the row disappears, the snackbar shows "Restored 1 item" (or "Restored *N* items") and a full re-scan starts. For a contact sheet only that burst shot is restored; the sheet image stays.

RULES & LIMITS

- Window: 30 days (30 × 24 h). Shrink rows are dated by when the copy was made; a shrink whose original is still in the trash stays listed (with Undo) however old it is.
- Rows deleted for good (Free this space now, Send the trash to computer) stay listed without Undo.
- Cancelling Android's dialog changes nothing. The restore is recorded before the dialog and resolved on the next start if the app is killed meanwhile.

WHAT THE USER SEES

Empty: "Nothing changed in the last 30 days."

SAFETY NOTES

A shrink Undo never deletes a copy unless the original is verifiably back and unchanged.

Code: `ui/RecentChangesScreen.kt`, `history/RecentChanges.kt`, `history/History.kt`

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6 Clean up

6.1 Clean up lists ("Suggestions")

PURPOSE

Suggest items worth deleting, sorted into lists. Nothing is deleted by the app on its own: the user selects and confirms in Android's trash dialog, and items stay restorable for about 30 days.

WHERE

Home › **All tools** › **Suggestions**. Screen title "Clean up". It opens on "Old screenshots".

HOW IT WORKS

- A scrolling row of list buttons "*list title* · *N*" (the count is missing until the list is worked out), in fixed order, then a "Duplicates" button that opens the Duplicates screen. The selected list's description is shown underneath.
- Thumbnail grid; tap a tile to select (red border and check). Caption under a tile: "*size* · *issues*" (e.g. "2 MB · Blurry, Too dark").
- "Select all" / "Clear" apply to the current list.
- "Review one by one" opens the swipe screen over the current list (see Swipe review); when finished, the swipe decisions replace that list's selection — nothing is trashed yet. Back during swiping keeps the old selection.
- Bottom bar: "Frees *size* once the trash empties · restorable for about 30 days" and "Move *N* to trash (*size*)" (current list's selection; disabled when nothing is selected or a dialog is open). Android's trash dialog follows.

RULES & LIMITS

List	Description shown	Who appears
"Old screenshots"	"Screenshots taken more than 30 days ago"	Photos (not screen recordings) whose album name contains "screenshot" or whose file name starts with "screenshot" (any case); age ≥ 30 days
"Old chat media"	"Photos and videos saved from chat apps, older than 30 days"	Album name contains one of the words whatsapp, telegram, snapchat, messenger, signal, instagram, wechat, line; or a WhatsApp-style file name (IMG/VID/AUD/PTT-*8 digits*-WA*digits*); screenshots excluded; age ≥ 30 days
"Low-quality photos"	"Blurry, too dark, overexposed or tiny photos"	Photos of any age with at least one issue (below)
"Expired"	"Items you chose to keep only until a date that has passed"	Any item whose "keep until" date has passed (set by the user or by the expiry reader)
"Old receipts & tickets"	"Bills, receipts and tickets older than 30 days (ID documents are never shown)"	Photos whose text reads as a receipt/bill or a ticket/pass; age ≥ 30 days
"Greetings & forwards"	"Good-morning cards, festival wishes and quote images, older than a week"	Photos whose text reads as a greeting (from any album); age ≥ 7 days

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List	Description shown	Who appears
"Old downloads"	"Photos and videos in your Download folder, older than 30 days"	Album named exactly "Download" or "Downloads" (any case); age ≥ 30 days

Low-quality issues (photo decoded at about 256 px):

Issue label	Rule
"Very small"	Original's long edge 1–639 px
"Too dark"	Average brightness below 35 and 95th-percentile brightness below 90 (0–255 scale)
"Overexposed"	Average brightness above 225, or more than 40 % of pixels at 250 or above
"Blurry"	Not "Too dark", pixel variance at least 300 (plain sky or walls are not flagged), and sharpness (Laplacian variance) below 100

- Favourites never appear in any list. Items with a future "keep until" date are hidden from every list except Low-quality photos.
- Each list is sorted largest file first. No list uses the "protect new photos" window; the ages above are the only time limits.
- Nothing is selected by default. A selection is shared across lists (an item in two lists shares its tick), but the button trashes only the current list's selected items.
- Videos are not text-checked and appear immediately. Photos appear only after the ID shield has checked them; items that are probably in a list are checked first, in batches of 24, then the low-quality scan runs, then all other photos (this fills receipts and tickets).

WHAT THE USER SEES

- While checking: a progress bar and "Checking for ID documents… *N* of *N*. More items appear as they're checked." (the low-quality scan shows its own count in the same caption, so the counter can jump).
- Empty list while checking: "Checking your photos first. ID documents are kept out of every list, so items appear here as they're checked." Empty list when done: "Nothing here — nice."
- On OK in Android's dialog: the items leave every list, are logged in Recent changes and in Your month; no message. Cancel: nothing changes, selection kept. Items no longer in the gallery are dropped; if none remain nothing happens.

SAFETY NOTES

If preparing the lists fails, all lists stay empty (fails closed). Nothing moves without Android's dialog.

> **DOCS VS CODE**
> The round-2 plan says the greetings list is for chat-app images; the code includes greetings from any album. "Old downloads" includes only albums named exactly Download/Downloads (not differently named subfolders).

Code: `ui/CleanupScreen.kt`, `cleanup/CleanupLanes.kt`, `cleanup/LowQualityFinder.kt`, `shrinkflow/CleanupViewModel.kt`

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6.2 ID shield

PURPOSE

Identity documents, and photos the app could not read, are never suggested for removal: not in the Clean up lists, Duplicates, Quick review or Chat media.

WHERE

Runs automatically behind Clean up, Duplicates, Quick review and Chat media. Visible through the checking captions, the receipts list's description, the "Not checked yet" label in Duplicates and the not-checked line in Chat media.

HOW IT WORKS

On-device text recognition (Latin script) reads each photo (decoded at 1280 px). The text is classified:

Kind	Rule
ID document (shielded)	Contains any of: aadhaar, unique identification authority, government of india, income tax department, permanent account number, passport, republic of india, driving licence/license, election commission, voter, national id, identity card, social security, driver's license/licence, pan card, date of birth, valid till, issuing authority; or an Aadhaar-style number (dddd dddd dddd), a PAN (AAAAA9999A) or a passport number (A1234567). ID wins even if it also looks like a receipt.
Ticket or pass	At least 2 ticket words (boarding pass, pnr, gate, seat, departure, e-ticket, ticket, booking id, platform, coach, flight, show time, screen) and at least as many ticket words as receipt words
Receipt or bill	At least 2 receipt words (total, subtotal, gst, cgst, sgst, invoice, receipt, bill no, amount, paid, payment successful, upi, txn, transaction id, order id, qty, rs., ₹, tax, cash)
Greeting or forward	Any one greeting phrase in English or romanised Hindi (e.g. good morning/night, shubh prabhat, happy diwali/holi, eid mubarak, jai shri, quote of the day, happy *weekday*). Devanagari-only cards are not detected.
None	Blank text, or nothing matched
Unknown (shielded)	The photo could not be decoded or text recognition failed; not remembered, so it is retried next session

RULES & LIMITS

- Results are remembered per photo and file size; a changed file is read again.
- Photos not yet checked never show in any Clean up list, are never pre-selected in Duplicates (shown as held, “Not checked yet”), never enter Quick review, and are never offered in Chat media. Videos need no check.

WHAT THE USER SEES

“(ID documents are never shown)” in the receipts description, and the checking captions above.

SAFETY NOTES

Fails closed everywhere: unreadable means protected, unchecked means held.

Code: `cleanup/DocumentClassifier.kt`, `cleanup/OnDeviceVision.kt`

6.3 Expiry dates and "Keep for a while"

PURPOSE

Give tickets, OTP screenshots and user-chosen items a "keep until" date; after it passes they appear in “Expired”.

WHERE

- Automatic (expiry reader): runs when a photo is first read by the ID shield.
- Manual: Clean up list → select items → row “Keep selected for” with “1 week”, “1 month”, “3 months” (shown when at least one item in the current list is selected).

HOW IT WORKS

- **OTP / verification code**: text contains otp, one time password / one-time password, verification code, security code or login code, **and** a 4–8 digit number → expires exactly 24 h after the photo was taken.
- **Ticket**: among the dates printed on it (numeric day-first, yyyy-mm-dd, "12 Mar 2026", "Mar 12, 2026"), the latest date between 7 days before and 400 days after the photo was taken is used; the ticket expires at the start of the next day. No such date → no expiry.
- **Keep selected for**: sets keep-until = now + 7, 30 or 90 days on the selected visible items (replacing any earlier date). They leave the lists and are unselected; no message.

RULES & LIMITS

- The reader never overwrites an existing keep-until date and does nothing when the date taken is unknown or the photo is an ID or unreadable.
- A ticket for a future trip is hidden from Old receipts & tickets until the day after travel, then shows in Expired (and in receipts too if it is 30+ days old).

WHAT THE USER SEES

Items reappear in “Expired” after their date. In Quick review, Expired items are always pre-marked for trash.

SAFETY NOTES

An expiry only makes an item a suggestion; it is never deleted automatically.

> **TESTER NOTE**
> For Low-quality photos, "Keep selected for" hides them only until the app is restarted; the low-quality scan does not read keep-until dates.

> **DOCS VS CODE**
> Design v2 says tickets expire the day after "the printed travel date". The code uses the latest plausible printed date within −7 to +400 days of capture.

Code: `cleanup/ExpiryReader.kt`, `cleanup/ExpiryTags.kt`, `ui/CleanupScreen.kt`

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6.4 Swipe review

PURPOSE

Decide one item at a time with a swipe. Used by Clean up ("Review one by one"), Quick review and Chat media.

WHERE

See those three features.

HOW IT WORKS

- Header "Review *N* of *M*" and "Space you'll free: *size*" (total marked for trash).
- Under the photo: "*reason* · *file name* · *size* · *issues*" (reason only in Quick review and Chat media).
- Buttons "Keep" and "Trash"; caption "Swipe right to trash, left to keep. You'll review the selection before anything moves."
- Drag right past 220 px to trash, left past 220 px to keep; less springs back. An overlay "Trash"/"Keep" appears at 110 px.
- Pre-marked items can be swiped back to keep.

RULES & LIMITS

Each decision moves to the next item.

WHAT THE USER SEES

As above.

SAFETY NOTES

Swiping alone never trashes anything; a confirmation step follows (in Chat media it is Android's dialog directly).

Code: `ui/CleanupScreen.kt`

6.5 Quick review

PURPOSE

One swipe queue over every Clean up list plus duplicate and burst extras, ending in a single trash confirmation.

WHERE

- **Home** › **Get space back** → "Clear the junk" → "Start swiping".
- **All tools** › **Quick review** ("Swipe left to keep, right to clear").
- Weekly notification "Your weekly tidy-up is ready"; Make room's review step.

HOW IT WORKS

1. Summary: "*N* suggestions" and "Up to *size* you could free. Swipe right to trash, left to keep." (+ "More appear as photos are checked." while checking). Button "Start swiping".
2. Start swiping freezes the queue; items found later are not added in this visit.

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3. Swipe through the queue. **Back during swiping leaves Quick review and discards the decisions.**
4. After the last item: if anything was marked, "*size*", "from *N* items you swiped to trash", button "Move *N* to trash", "They stay restorable in the trash for 30 days." and "Review again"; otherwise "You kept everything." and Review again.
5. Move N to trash → one Android trash dialog. OK trashes and logs each item with its reason; Cancel changes nothing.

RULES & LIMITS

- Queue order: list items in list order (Old screenshots, Old chat media, Low-quality photos, Expired, Old receipts & tickets, Greetings & forwards, Old downloads; each largest first), then the removable extras of duplicate groups, ordered exact copies → bursts → very similar. Held members (not checked yet, recent, favourite, Never-touch album) and keepers never enter the queue. An item appears once; the first reason wins.
- Reasons shown: the list title, "Exact copy", "Burst, not the best shot", "Very similar shot".
- Pre-marked for trash: Expired items always; Old screenshots, Greetings & forwards and Low-quality photos when the matching standing rule is on. Duplicate extras are not pre-marked.
- Opening Quick review starts list preparation and the duplicate scan if they have not run yet.

WHAT THE USER SEES

Empty: "Still checking your photos. Suggestions appear here as they're ready." while checking; "Nothing to review. Your gallery is tidy." when done.

SAFETY NOTES

One Android dialog for everything marked; nothing moves before it.

> **TESTER NOTE**
> After a successful trash the summary shows "You kept everything." (no success message). "Review again" reuses the frozen queue, which still contains the items already trashed.

> **DOCS VS CODE**
> Design v2 says standing rules pre-select. In code Expired items are pre-selected even with no rules on.

Code: `ui/QuickReviewScreen.kt`, `shrinkflow/QuickReviewQueue.kt`, `cleanup/StandingRules.kt`

6.6 Duplicates: exact copies, bursts, very similar shots

PURPOSE

Find copies and near-copies, suggest the best one to keep, and trash the rest after confirmation.

WHERE

All tools › **Suggestions** → last button "Duplicates". Screen title "Duplicates". The scan runs once per app session.

HOW IT WORKS

- Header: "*N* groups · the sharpest copy of each is kept — long-press a photo to keep it instead".

- Each group card: label "Exact copies", "Burst · best shot kept" or "Very similar"; every copy shown, 4 per row. The keeper's caption is "Keep" (green); others show their size; selected copies have a red border and a check.
- Only **removable** members are pre-selected: checked for ID documents, not shielded, not the keeper and not **held**. Held members stay visible with a label: "Not checked yet", "Protected · favourite", "Protected · recent" (inside the protect window), "Never-touch album", or "Album rules unavailable" (album rules could not be read). Held members are never pre-selected but can be ticked by hand. As unchecked photos get checked, their label clears (they stay unselected).
- Tap a non-keeper to toggle; tapping the keeper does nothing. **Long-press** any photo to make it the keeper; the old keeper becomes selected only if it is removable (not held).
- Bottom bar: "Frees *size*" (when something is selected) and "Move *N* duplicates to trash" (disabled when nothing is selected). OK in Android's dialog removes the items and drops groups left with fewer than 2.

RULES & LIMITS

Group type	Rule
Exact copies (photos and videos)	Same byte size → same quick hash (first and last 64 KB plus length) → same full SHA-256. Unreadable files never join a group. Hashes are remembered per file id, size and date modified.
Burst (photos not in an exact group)	Same album, date taken known, consecutive shots at most 1.5 s apart, at least 3 shots
Very similar (remaining photos)	9×8 grayscale difference hash; grouped (transitively) when at most 10 of 64 bits differ

- **Keeper for bursts and very-similar groups**: favourite → best eyes-open/smile score (no faces scores lowest; the worst face counts) → sharpness → resolution → newest date modified.
- **Keeper for exact groups**: favourite → resolution (0 for videos) → newest date modified.
- ID documents and unreadable photos are removed from groups; a group with fewer than 2 left disappears. In an exact-copy group the copies are byte-identical, so one ID or unreadable copy removes the whole group, and one checked copy counts as checking all of them.

WHAT THE USER SEES

- Scanning: a spinner with "Comparing files… *N* of *N*", "Looking for similar shots… *N* of *N*" or "Preparing…".
- No groups: "No duplicates found — nice and tidy."
- Recent changes logs "Duplicate or burst extra".

Contact sheet (burst groups only)

- Offered only for burst groups where nothing besides the keeper is held (all shots checked, none recent, favourite or in a Never-touch album). Link "Keep best + contact sheet" with caption "Saves one image with all *N* shots, keeps the best in full, and moves the other *N−1* to the trash."
- Makes a grid JPEG (480 px cells; 2 columns up to 4 shots, 3 up to 9, otherwise 4; the best shot framed in green; footer "*N* shots · *date/time* · best shot kept in full"), saved as `Pictures/GallerySlim/Contact sheets/Burst_`*best name*`_sheet.jpg` (JPEG quality 88, dated like the best shot).

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- Then **all shots except the keeper** go into one Android trash dialog, whatever their ticks (none of them can be held, by the rule above).
- OK: "Kept the best shot and a contact sheet of all *N*" (logged as "Folded into a contact sheet", reason Burst). Cancel: the sheet image is deleted so nothing is half-done. Failure: "Couldn't make the contact sheet".

SAFETY NOTES

The keeper can never be selected for trash. Held members (unchecked, recent, favourite, Never-touch) are never pre-selected.

> **DOCS VS CODE**
> Design v2 gives one keeper order for all groups; in code exact groups skip the face and sharpness steps, and "newest" means date modified.

Code: `cleanup/DuplicateFinder.kt`, `cleanup/PerceptualHash.kt`, `cleanup/ContactSheet.kt`, `ui/DuplicatesScreen.kt`

6.7 Chat media

PURPOSE

Show how much space chat apps' photos and videos take, and review old ones.

WHERE

Get space back › **Clear the junk** › **Chat media**, or **All tools** › **Chat media**. Screen title "Chat media".

HOW IT WORKS

- Top line: "Chat apps keep *size* of photos and videos on this phone."
- "Older than" buttons: "1 month", "3 months" (default), "6 months", "1 year".
- One row per album (folder), largest first: name, size, a bar for its share, and "*N* photos · *N* videos · *size* older than the cut-off". A blank album name shows as "Chat media".
- If old chat photos are still unchecked: "*N* photos (*size*) not checked for ID documents yet — they appear here once checked."
- Button "Review *N* old items (*size*)" when at least one offered item exists, with "Swipe to choose; nothing moves until you confirm, and the trash keeps them 30 days."
- The swipe screen starts with nothing marked, largest first, reason = album name. **After the last swipe, Android's trash dialog opens immediately** for the items marked. Back during swiping discards the decisions.

RULES & LIMITS

- Chat items: same rule as the "Old chat media" list (chat-app words in the album name, or WhatsApp-style file names).
- A month counts as 30 days, so "1 year" means 360 days or older.
- Offered items: older than the cut-off, checked for ID documents (videos need no check), not an ID or unreadable photo, not a favourite, not inside the protect window, not in a Never-touch album (nothing is offered if album rules cannot be read). Held items still count in the folder totals. Keep-until dates are not checked.

WHAT THE USER SEES

OK trashes the items (logged with the album name as reason); Cancel changes nothing.

SAFETY NOTES

Android's dialog is the confirmation. The ID shield applies: unchecked photos are held back until checked.

> **DOCS VS CODE**
> The round-3 plan describes space per chat folder *and month*; the code shows per folder only. The swipe caption says "You'll review the selection before anything moves", but Chat media goes straight to Android's dialog.

Code: `ui/ChatDietScreen.kt`, `cleanup/ChatDiet.kt`

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7 Keeping it slim

7.1 Album rules

PURPOSE

Choose, per album, how its photos are treated when shrinking: protect the album completely, or make lighter copies of its old photos.

WHERE

Home › **All tools** › **More** › **Album rules**. Header “Album rules”; intro “Choose how each album is treated when you shrink by hand and overnight. "Never touch" leaves every photo and video in the album exactly as it is.”

HOW IT WORKS

1. Every album found by the scan is listed: most items first, then name A–Z. A blank name shows as “Unnamed album”. Each row: name, “*N* items”, and the current rule (grey for Default, warning colour for Never touch, brand colour for Light copy).
2. Tap a row → dialog titled with the album name, radio options: “Default”, “Light copy for photos older than 1 year”, “Light copy for photos older than 2 years”, “Light copy for photos older than 3 years”, “Light copy for photos older than 5 years”, “Never touch”.
3. Choosing an option saves it at once and closes the dialog; “Close” dismisses without change.

RULES & LIMITS

- "Older than N years" = age above N × 365.2425 days; unknown age is never "old".
- **Light copy rule**: only still photos older than N years get the Light copy preset (2048 px); newer photos and all videos in the album get the global preset. The rule itself does not exclude favourites; they are kept out by the protection defaults (an old favourite ticked by hand would get a light copy).
- **Never touch** removes the album's items from manual shrinks, Big videos light copies, Trim, Make room, overnight and trips. It does not apply to the motion-photo strip.
- **Precedence**: Never touch beats everything. An album's Light-copy rule beats the global cooling setting for that album. Big videos' forced Light copy ignores Light-copy album rules but still honours Never touch.

WHAT THE USER SEES

- Empty: “No albums found yet. Run a scan first.”
- When items were dropped: “Left *N* untouched — their album is set to Never touch”; for Trim: “This video's album is set to Never touch”.

SAFETY NOTES

A rule only picks a preset or excludes items; it never deletes anything.

> **DOCS VS CODE**
> The round-2 plan says only overnight and manual shrink honour album rules. The code also applies them to Make room, trips, light copies and Trim.

Code: `shrinkflow/AlbumRules.kt`, `ui/AlbumRulesScreen.kt`

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7.2 Cooling: lighter copies of old photos

PURPOSE

For albums left on Default, photos older than a chosen age become 2048 px light copies instead of normal shrinks.

WHERE

Settings › **Automatic** › **Lighter copies of old photos**: chips “Off”, “Older than 1 yr”, “Older than 2 yr”, “Older than 3 yr”; caption “2048 px copies. Favourites never; album rules win; originals wait in Ready to clear.” Also set by the fifth standing-rule question (Yes = 2 years, No = Off).

HOW IT WORKS

When a photo is shrunk (manually, by Make room, overnight or on a trip), it gets the Light copy preset if it is a still photo, not a favourite, not a motion photo, not Ultra HDR, older than the chosen years, and in a Default album.

RULES & LIMITS

- Choices 0 (off), 1, 2, 3 years. Default: Off.
- Not applied when Big videos forces Light copy.

WHAT THE USER SEES

No separate screen; the copies appear in Ready to clear like any shrink.

SAFETY NOTES

Originals still wait in Ready to clear.

Code: `shrinkflow/AlbumRules.kt`, `ui/SettingsSheet.kt`

7.3 Storage floor: "Keep this much space free"

PURPOSE

A free-space goal that overnight shrinking works toward and the morning note reports on.

WHERE

Settings › **Automatic** › **Keep this much space free**: chips “Off”, “5 GB”, “10 GB”, “15 GB”, “20 GB”; caption “Overnight shrinking works toward this goal, and the morning note tells you what to clear.” Also set by the fourth standing-rule question (Yes = 10 GB, No = Off).

HOW IT WORKS

- Gap = goal – current free space (binary GB), never below 0. If free space cannot be read, no gap is assumed.
- Overnight, when below the goal, items with the largest estimated saving go first (Make room rungs 1–2); the night's caps (300 photos / 10 videos) are then filled as usual.
- Make room offers a target chip "Back to N GB free" and uses the gap as its default target.

RULES & LIMITS

- Default: Off. Values outside 0/5/10/15/20 are stored as Off.

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- The floor has an effect only through "Shrink overnight" (off by default) and Make room. Answering Yes to the floor question does not switch overnight shrinking on.

WHAT THE USER SEES

Morning note (channel "Overnight shrink results"), title "Below your free-space goal" when still below, otherwise "Overnight shrink finished". Text:

Situation	Text
Above the goal, items waiting	"Above your *N* GB goal. *N* original(s) ready to clear (*size*)."
Above the goal, nothing waiting	"Above your *N* GB goal."
Below, items waiting	"You're *X* below your *N* GB goal. Clear *N* original(s) in Ready to clear to get *size* back (space returns when the trash empties)."
Below, nothing waiting	"You're *X* below your *N* GB goal. Open Make room to find more space."

"original(s)" is singular for 1. Tapping the note opens Ready to clear (even when the text mentions Make room).

SAFETY NOTES

The goal never causes deletion; it only orders shrinking and informs.

> **DOCS VS CODE**
> Design v2 and the round-3 plan say the overnight job "sizes its batch to the floor gap". The code puts gap-closing items first and still fills the full 300/10 caps.

Code: `shrinkflow/RoomPlanner.kt`, `work/OvernightWorker.kt`, `work/ShrinkNotifications.kt`

7.4 Make room

PURPOSE

A plan to free a chosen amount, gentlest steps first (the "ladder of regret"). It never deletes anything.

WHERE

- **All tools** › **Your phone** › **Free up space**.
- The low-space notification. Header "Make room".

HOW IT WORKS

1. Intro: "The gentlest steps come first. Shrinking makes new, smaller copies; the originals wait in Ready to clear, and space comes back once the trash empties."
2. "How much room?" chips: "Back to *N* GB free · *X*" (only when a storage goal is set and the gap is above 0), "2 GB", "5 GB", "10 GB". Default target: the gap if there is one, else 2 GB. The pick sticks for the app session.
3. The plan is built from up to four rungs in this order, stopping as soon as the running estimate reaches the target (later rungs may not appear). Each card shows "about *X*" and an on/off switch (all on whenever the plan is rebuilt).
4. Bottom: "About *X* back" and the main button (table below).

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5. If rung 1 or 2 is on, one shrink batch starts and the progress screen opens; if rung 3 or 4 is on, Quick review then opens on top (with its normal full queue, not only the plan's items).

RULES & LIMITS

#	Card title	Blurb	Rule
1	"Shrink *N* photos" / "Shrink *N* videos" / "Shrink *N* photos and *M* videos"	"Smaller copies that look the same. The originals wait in Ready to clear."	JPEG, PNG and H.264 with an estimated saving (Ultra HDR not included), largest saving first
2	"Lighter copies of *N* photos from *YYYY*" (or a year range)	"2048 px copies of older photos: fine on a phone, softer if you zoom or print. Favourites are never included."	Only if rung 1 falls short. Still JPEG/PNG photos, not favourite/motion/Ultra HDR, older than: the album's Light-copy years; else the cooling years; else 2 years. Estimate 70 % of size; a photo already in rung 1 moves here and only its extra saving counts
3	"Clear *N* expired items"	"Their keep-until date has passed. You check them in Quick review first."	The Expired list
4	"Review *N* suggestions"	"Old screenshots, chat media, duplicates and more. You choose in Quick review."	The Quick review queue without Expired

- All four rungs leave out items already in Ready to clear, items in Never-touch albums and protected items (favourites and photos inside the protect window). Rungs 3–4 also skip zero-size items and never repeat an item; largest first.
- Sizes in Make room, the storage-goal note and the trip note use "*1.2* GB" from 1 GB, whole MB below that, and whole KB (at least 1 KB) below 1 MB.
- The lists fill in progressively, so the plan may change while the screen is open.

Switches on	Main button
Rung 1/2 and rung 3/4	"Shrink *N*, then review the rest"
Only rung 1/2	"Shrink *N* items"
Only rung 3/4	"Review in Quick review"
Nothing	"Nothing selected" (disabled)

N here has no thousands separator.

WHAT THE USER SEES

- Empty plan: "Nothing to suggest right now." Plan short of the target: "That's everything we found — about *X* of *Y*."
- Vault on without folder: "Pick a vault folder first" and nothing runs.

SAFETY NOTES

Rungs 1–2 only make copies (originals wait in Ready to clear); rungs 3–4 only open review.

Code: `ui/MakeRoomScreen.kt`, `shrinkflow/RoomPlanner.kt`

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7.5 Trip mode

PURPOSE

"I need N GB free by a date": a one-off job the evening before shrinks to close the gap.

WHERE

Make room → bottom card "Going somewhere? Make room by a date" (tap to expand).

HOW IT WORKS

1. "When": "In 2 days", "This weekend" (default), "Next week". "Free space I'll need": "2 GB", "5 GB" (default), "10 GB". Button "Plan it".
2. Snackbar: "We'll make room the evening before *Sat 24 Oct*". The card then shows "Making room by *Sat 24 Oct* · *5* GB free", "We'll shrink the evening before. The originals will wait in Ready to clear." and "Cancel trip" (removes it without a message).
3. Two stages: a kick-off job fires at the run time below (its only condition is battery not low, which plugging in satisfies), then starts the actual run as a priority job; if Android stops it, it runs again and continues where it left off (finished items are already queued). If another batch is running, it retries after 15 minutes.
4. At run time: gap = needed GB (binary) − free space. If enough space is free, it only notifies. Otherwise it runs Make room rungs 1–2 with the overnight caps (300 photos / 10 videos).

RULES & LIMITS

- Dates: "In 2 days" = today + 2; "This weekend" = today if Saturday or Sunday, else the coming Saturday; "Next week" = the coming Monday.
- Runs at 20:00 local time the day before the trip date, or immediately if that time has passed (e.g. "This weekend" chosen on a Saturday runs at once). No charging or idle condition; it waits only while the battery is low and unplugged.
- One trip at a time; a new plan replaces the old one. The trip disappears after its day ends or after the job has run.
- Light copies only for photos the user opted into (cooling on, or an album Light-copy rule); the 2-year fallback is not used.
- Vault on without folder: posts "Overnight shrink skipped" and drops the trip.

WHAT THE USER SEES

Progress notification "Shrinking your gallery", then (channel "Overnight shrink results"; tap opens Ready to clear):

- "Ready for your trip" / "You already have *N* GB free. Nothing to do."
- "Trip prep: *N* items shrunk" or "Trip prep finished", with "Clear *N* originals in Ready to clear to get *X* back (space returns when the trash empties)." (+ " Open Make room to review suggestions for the rest." when X is below the gap), or "Nothing more could be shrunk. Open Make room to review suggestions."

SAFETY NOTES

A trip never deletes anything; it only queues items for Ready to clear.

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TESTER NOTE

"Trip prep: 1 items shrunk" is possible (no singular). The count in the text includes everything waiting in Ready to clear, not only this trip's items.

DOCS VS CODE

Design v2 says trips make light copies "only when cooling is on". The code also makes them for albums with a Light-copy rule.

Code: `work/TripWorker.kt`, `ui/MakeRoomScreen.kt`

7.6 Overnight shrink

PURPOSE

Shrink while the phone charges, so the user only has to review in the morning.

WHERE

Settings › **Automatic** › **Shrink overnight** (switch), caption "Shrinks while the phone is charging at night. Nothing is deleted. You review the results in the morning."

HOW IT WORKS

1. Turning it on asks for notification permission (Android 13+) and schedules a periodic job; turning it off cancels it.
2. Each run: a fresh read-only scan; eligible items are JPEG photos, PNG images and H.264 videos that are not protected, not in Never-touch albums and not already waiting in Ready to clear (Ultra HDR and motion photos excluded).
3. Preset: global, with album rules and cooling. The vault is used when on.

RULES & LIMITS

- Every 24 h, only while **charging**, **device idle** and **battery not low**. No clock time is enforced: "overnight" means whenever these hold.
- Caps: **300 photos and 10 videos** per run. Without a storage goal the order is the media library's order; with a goal, gap-closing items first.
- Vault on without a folder: the run is skipped.
- If a manual or trip batch is running, that night's run is skipped (one batch at a time).
- Default: off.

WHAT THE USER SEES

- Progress: "Shrinking your gallery" / "*done* of *total* · *name*" or "Scanning…".
- Morning (no goal set): "Overnight shrink finished" / "Ready to clear: *N* originals · *size*. Tap to clear them.", or "Overnight shrink — nothing to do" / "Your gallery is already slim." (also when items were tried but none succeeded). With a goal: see Storage floor. Tap opens Ready to clear.
- Vault problem: "Overnight shrink skipped" / "The vault is on but no vault folder is set. Pick one to resume overnight shrinking." (tap opens Home).

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SAFETY NOTES

Only writes, records and queues; never deletes (confirmation dialogs cannot appear while the phone sleeps).

> **TESTER NOTE**
> Morning, goal and trip notifications open Ready to clear behind the app lock (if on).

> **DOCS VS CODE**
> Design v2 lists "charging + idle"; the code also requires battery not low.

Code: `work/OvernightWorker.kt`, `work/ShrinkNotifications.kt`

7.7 Standing rules: the five questions, and the weekly nudge

PURPOSE

Five yes/no answers become standing rules: what Quick review pre-selects, the storage goal, cooling, and a weekly reminder.

WHERE

- Get space back → "Keep it tidy by itself" → "Set up" (until setup is done).
- All tools card "Let GallerySlim tidy for you" (until setup is done).
- **Settings** › **Automatic** › **Your rules** ("Five questions that decide what gets suggested for clearing") — always.

HOW IT WORKS

- One question at a time. Header "Your rules · *N* of 5"; under the question its detail line and "You can change this any time in Settings. Rules only suggest; you always confirm."; buttons "No" and "Yes"; footer "Step *N*: answer once, and GallerySlim handles the rest."
- Each answer is saved immediately. After the 5th: "Rules saved. Change them any time in Settings." and the screen closes. Going back part-way keeps the answers given, but setup is not marked done.

RULES & LIMITS

#	Question	Detail line	Yes / No
1	"Clear screenshots after a month?"	"Old screenshots are pre-selected in Quick review."	Pre-select Old screenshots / rule off
2	"Let good-morning cards and forwards go after a week?"	"Greeting and quote images from chats are pre-selected."	Pre-select Greetings & forwards / off
3	"Clear blurry and too-dark shots?"	"Low-quality photos are pre-selected."	Pre-select Low-quality photos (all four issues) / off
4	"Always keep 10 GB free?"	"Overnight shrinking works toward this goal."	Storage goal 10 GB / Off
5	"Lighter copies of photos older than 2 years?"	"Favourites and albums you protect are never touched."	Cooling 2 years / Off

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- All rules are off by default. Answering 4–5 overwrites the values set in Settings.
- **Weekly nudge**: every 7 days (first after 3 days), posted only if rule 1, 2 or 3 is on (it does not check whether anything matches). Channel "Weekly tidy-up" (silent): "Your weekly tidy-up is ready" / "Your rules picked what to clear. Take a look, then confirm once."; tap opens Quick review.

WHAT THE USER SEES

Pre-selected items in Quick review; the notification above.

SAFETY NOTES

Rules never delete; they only pre-select, and the user confirms in Quick review and Android's dialog.

> **DOCS VS CODE**
> The round-3 plan says a first-run card; the code offers the questions through the flow step, the All tools card and Settings, not at first launch. Question 4's detail line promises overnight work, but overnight shrinking stays off unless turned on separately.

Code: `cleanup/StandingRules.kt`, `ui/RulesSetupScreen.kt`

7.8 Low-space alert

PURPOSE

Warn once a day when the phone is nearly full.

WHERE

A notification; tap opens Make room.

HOW IT WORKS

Checked by the 12-hour free-space sampling job (not when the app opens).

RULES & LIMITS

Fires when free space is below 2 GB (2,147,483,648 bytes) and the last alert was at least 24 h ago.

WHAT THE USER SEES

Channel "Low space": "Running low: *2.1* GB free" (decimal GB, one decimal) / "Make room now, before the camera says it's full?"

SAFETY NOTES

Informational only.

> **DOCS VS CODE**
> The round-3 plan's wording "Running low. Make room now?" differs from the code's text above.

Code: `storage/StorageWidget.kt`

7.9 Home-screen widget and forecast

PURPOSE

Show free space and when the phone will be full, on the home screen.

WHERE

Android widget picker → GallerySlim widget ("Free space and when your phone will be full"). Minimum 180 × 110 dp, default 3 × 2 cells, resizable.

HOW IT WORKS

- Shows "GallerySlim", "*12.9* GB free" (decimal GB), a used-space bar, a forecast line and a button "Shrink now".
- Tapping the widget or the button only opens the app at Home (no shrink starts).

RULES & LIMITS

Forecast	Line shown
No forecast yet	"Tracking how fast it fills up"
0 days	"Almost full"
Above 365 days	"Plenty of room at this rate"
Otherwise	"Full in about *N* days at this rate" (no singular)

- Samples: on app start, widget update and every 12 h; at most one per 6 h; the last 120 kept.
- Forecast: samples from the last 30 days; at least 3 samples spanning at least 2 days; a least-squares line; free space must be falling. "Full" = below 500 MB. Days = (fitted free now − 500 MB) ÷ daily drop, rounded down, minimum 0.
- Refresh: every 6 h by the system, by the 12-hour job, and after deletions via Clean up, Duplicates, Quick review or Chat media (not after shrinks).

WHAT THE USER SEES

The same forecast also appears on Home (" · full in about *N* days") and in All tools (" · full in about *N* days at this rate"), hidden above 365 days.

SAFETY NOTES

Read-only.

Code: `storage/StorageForecast.kt`, `storage/StorageWidget.kt`

7.10 Your month (monthly report)

PURPOSE

A monthly summary of space freed, as a card to share.

WHERE

All tools › **More** › **Your month**, or the monthly notification. Header "Your month".

HOW IT WORKS

- Starts on the current month. "‹ Earlier" is always shown (no lower limit); "Later ›" only when an earlier month is shown.
- Card image, then one line: "Nothing freed in *Month Year* yet." or "Shrinking saved *X* across *n* items, and clearing removed *m* items (*Y*). Space from originals in the trash is counted when you confirmed them; it frees fully when the trash empties."
- "All time: *size* freed for real, *size* still waiting in the trash."
- "Share this card" (enabled only when the month's total is above zero) → share sheet "Share your month" with `gallery-slim-yyyy-MM.png` and the text "I freed *X* on my phone this month with GallerySlim".

RULES & LIMITS

- Shrinks counted: copies confirmed now whose creation time falls in that month; saving = original – copy.
- Cleared: a per-month running tally added at the moment of trashing (Clean up, Quick review, Duplicates, contact sheets, Chat media, "Move sent files to trash"); never reduced by a later undo; trash-export deletions are not counted again.
- Total = shrink saving + cleared. "Room for about N more photos" = total ÷ 3,000,000 bytes.
- Card: 1080 × 1080 PNG with the lines "My *Month Year* with GallerySlim", "Freed *X*", "*n* photos & videos made smaller", "*m* unwanted items cleared", "Room for about *N* more photos", "Done on my phone. Nothing uploaded." (decimal units).

WHAT THE USER SEES

Notification (daily check; posts once per month for the previous month on its first run after the month changes; skipped when nothing was freed): channel "Monthly report", "Your *Month Year*: *X* freed" / "Room for about *N* more photos. Tap to see and share." → opens Your month.

SAFETY NOTES

The card contains only numbers; sharing is user-initiated.

> **TESTER NOTE**
> The notification is about last month but opens the current month; tap "‹ Earlier" to see it. Shrinks are counted in the month the copy was made, not when it was confirmed.

Code: `report/MonthlyReport.kt`, `ui/ReportScreen.kt`

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8 Moving photos to a computer

Two ways exist. The **browser** way (protocol "gs1") needs nothing installed on the computer, but can only prove that the bytes left the phone. The **GallerySlim Receiver** way (protocol "gs2") re-reads every saved file from the computer's disk and proves it matches the phone's file ("verified"). Only verified files can ever be deleted for good.

8.1 Send to computer: the phone screen and the sharing service

PURPOSE

Copy chosen photos and videos from the phone to a computer on the same Wi-Fi. The phone runs a small local web server; a laptop browser or the Receiver pulls the files. Nothing is sent until the user scans a code shown on the computer and confirms on the phone.

WHERE

- Category screens: select items → "Send *N* to computer" (shown when at least one is selected, on every category including Motion photos).
- Ready to clear: header link "Send to laptop" (queues the originals of all waiting items).
- Send the trash to computer: "Send to computer first" (see below).

HOW IT WORKS

Screen "Send to computer" (back arrow; leaving the screen does **not** stop sharing). Subtitle "*N* items · *size*". The screen stays on while open.

- Receiver hint (once sharing has started): "Using the GallerySlim Receiver? Open it on the computer and scan its code. This phone's code is *K7-42*."
- **Step 1 — "On the computer, open this link"**: a big link `galleryslim.local:`*`port`* (when the phone's own name service runs) or *`phone IP`*`:`*`port`*; tap to copy the full http:// address. If the friendly name is shown: "If that name doesn't open: *ip:port*". Caption "Same Wi-Fi as this phone · tap to copy". Link "Send the link to my computer…" opens the share sheet ("Send the link"), caption "WhatsApp, email, Telegram… then click it on the computer." Before the link exists: the error text or "Starting…".
- **Step 2 — "Scan the code the computer shows"**: button "Scan the computer's code" (enabled when the link exists and the queue is not empty); caption "You'll confirm with your fingerprint or PIN. Only the screen you scan can receive."
- **Step 3 — "Save on the computer"**: before approval, "The page offers one organized ZIP (sorted by month) or a folder of your choice."; after, a progress bar and "*sent* of *total* sent", then a check and "All *total* sent". Progress resets on each new approval.
- **Step 4 — "Free the space on this phone"**: shown once at least one queued file was delivered in this run (wording per way, below).
- Bottom: red "Stop sharing" stops the service and leaves the screen.

Scan screen: title "Scan the laptop's code", hint "Point at the square on the laptop screen". If the camera is denied: "The camera is only used to read the code on your laptop's screen." and "Allow camera". Only codes starting with `gs1:` or `gs2:` are accepted; others are ignored.

Confirmation: Android's biometric/PIN prompt titled "Send these files to your computer?" (subtitle per way, below). It always asks, even if the app was unlocked. On a phone with no screen lock and no

biometrics, an in-app dialog asks instead: title "Send these files?", the same subtitle text followed by "This phone has no screen lock, so anyone holding it could do this. Setting a screen lock makes sending safer.", buttons "Send" and "Cancel". Files are never sent without one of these confirmations. Cancel or error → snackbar "Not sent".

RULES & LIMITS

Rule	Value
Network	Wi-Fi only (not a VPN), IPv4. Otherwise "Connect the phone to Wi-Fi first".
Port	First free of 8765–8769, bound to the phone's Wi-Fi address only; random port if all five are busy (then the Receiver cannot find the phone by itself)
Friendly name	`galleryslim.local`, answered by the app's own multicast name responder
Who may connect	Only computers in the phone's own Wi-Fi subnet (or private addresses if the subnet is unknown); others are disconnected without a reply
Host check	Every request's Host header must be the phone's own Wi-Fi IP or `galleryslim.local` (with or without the port); otherwise it is answered "421 Misdirected Request" (blocks web pages that re-point their own name at the phone)
Connections	At most 8 at once, 5 per address; beyond that "503 Busy"
Request limits	Headers within 10 s; at most 64 header lines, 32 KB total, lines of at most 8191 characters (plus the line ending); POST bodies at most 16 KB; 30 s without progress closes a connection
Sessions	Expire after 10 min idle (each chunk sent keeps them alive); at most 32 unapproved; each pinned to the computer's IP address
Short code	2 letters (A–Z without I and O), "-", a number 10–99, e.g. "K7-42"; new each time sharing starts
Auto-stop	Wi-Fi lost; app swiped away; Android's time limit for this service (Android 15+); about 30 min with no approved session and no stream (checked every 60 s). In practice about 40 min after the last transfer, or about 30 min after start if nothing was approved
Stopping	Revokes all sessions, cuts transfers, and empties the queue
Power	While a file streams, the phone holds a CPU wake lock and a Wi-Fi lock

WHAT THE USER SEES

Ongoing notification "Ready to send to your computer" / "Open the link on the computer and scan its code · stops by itself after 30 min" with action "Stop sharing"; tap opens the app. Start failure: "Couldn't start sharing (notification blocked)".

SAFETY NOTES

Files go only to the computer whose code was scanned and confirmed; the server is LAN-only and same-subnet only. Files are streamed from the gallery; no copies are made on the phone.

> **TESTER NOTE**
> Without Wi-Fi (or on a VPN) step 1 shows "Connect the phone to Wi-Fi first" and a brief notification "Not sharing: no Wi-Fi" appears. The scan screen says "laptop" while the button says "computer".

Code: `ui/share/SendToLaptopScreen.kt`, `ui/share/QrScanScreen.kt`, `share/ShareService.kt`, `share/LanHttpServer.kt`, `share/ShareSession.kt`

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8.2 Send to computer via a browser (gs1)

PURPOSE

Save photos on any computer with just a browser, as one ZIP sorted by month or as single files.

WHERE

Send to computer screen, step 1 link opened in the computer's browser.

HOW IT WORKS

1. Each page load creates its own session and shows a QR code. Page (tab title "GallerySlim · Send to this computer"): heading "Send to this computer", "From *phone model* · only this screen can receive", the QR code, steps (1) "On the phone, tap" "Scan the laptop's code", (2) "Point the camera at this square.", (3) "Confirm with your fingerprint or PIN.", and "Waiting for the phone…".
2. Phone: "Scan the computer's code" → scan → prompt subtitle "*N* files (*size*) to the screen you just scanned" → confirm.
3. Snackbar: "Sending — pick where to save on the computer". Wrong or stale code: "That code isn't from this phone. Reload it on the computer and try again."
4. The page checks every 1.5 s; after approval it shows "Ready to save", "*n* files · *size*" and:
   - "Save all as one organized file" — hint "One ZIP in your Downloads folder. Unzip it: photos are sorted into **GallerySlim / year-month** folders."
   - "Save into a folder I choose" — hint "Sorted into year-month folders inside the folder you pick." Shown only when the browser supports folder access in a secure context, which the phone's plain-http page is not, so it is normally hidden.
   - "Download files one by one instead".
5. Expired session (while still waiting): "This link has expired" / "Reload the page on the laptop and scan the new code."

RULES & LIMITS

- **ZIP**: named `GallerySlim yyyy-MM-dd HH.mm.zip` (phone's time); no compression (DEFLATE at level 0); entries `GallerySlim/yyyy-MM/name` by date taken, or `GallerySlim/undated/name`; duplicate names become "name (2).ext"; entry time = date taken; slashes and control characters become "_".
- **ZIP button**: changes to "Preparing your file…", then after 1.5 s (a timer, not a real check) "Saved as one file ✓" and the heading "Check your Downloads folder".
- **One by one**: each file goes loose into Downloads under its original name (no month folders), 2 at a time; rows show "✓ saved" or "failed"; at the end "All saved ✓" or "*n* of *m* saved" and "Saved to your Downloads folder."
- **Folder picker** (if shown): `chosen/GallerySlim/yyyy-MM or undated/`, 4 at a time, " (2)" on clashes; "Saved into the folder you chose, sorted by month."
- **"Fully sent"**: a single file whose whole content was written in one response from byte 0, or a ZIP entry read to the end with the whole ZIP finished while sharing still ran. A cut-off ZIP marks nothing. This does not check the copy on the computer.

WHAT THE USER SEES

Step 4 on the phone: "*N* of *M* files (*size*) were fully sent. Open a few on the computer to check them, then move them to the trash here." Button "Move *N* sent files to trash" → dialog "Did the

files arrive?" / "Only files that were fully sent are included. They go to the phone's trash, where you can restore them for 30 days. Check that they open on your computer first." / "Yes, move to trash", "Not yet" → Android's trash dialog. OK: "Moved *N* sent files to the trash" (recorded in Moved to computer and Your month). Cancel: nothing changes. None left in the gallery: "These files are no longer in the gallery".

SAFETY NOTES

Browser sends are unencrypted on the local network and are never "verified"; they can move originals to the trash (30-day restore) but never delete them for good.

> **DOCS VS CODE**
> The phone's step 3 promises "a folder of your choice", but the folder option is normally hidden on the plain-http page. The ZIP "Saved as one file ✓" appears on a timer, before the download ends.

Code: `share/SharePage.kt`, `share/LanHttpServer.kt`, `ui/AppNavHost.kt`

8.3 Send to computer via the GallerySlim Receiver (gs2, phone side)

PURPOSE

Send to the Receiver program, which saves each file, re-reads it from its own disk and proves it matches; the phone then marks it **verified**.

WHERE

Same Send to computer screen; on the computer, open the Receiver (see GallerySlim Receiver).

HOW IT WORKS

1. The Receiver finds the phone (it shows the phone's name and short code) and asks for a session, sending its computer name, system and folder name.
2. Phone: scan the QR code shown by the Receiver (`gs2:…`). If it matches no waiting session: "That code has expired. Restart the receiver on the computer and scan again." (no prompt).
3. Prompt subtitle "*N* files (*size*) to *computer name*" → confirm → "Sending to *computer name*. Each file is checked on the computer."
4. For each file, the Receiver sends the SHA-256 and size of the copy it re-read from disk. The phone hashes the whole current file itself. If both match, the file is **verified**, saved immediately in the app's history (survives restarts), and counts as delivered. A mismatch is silently left out of the counts. A file deleted from the phone during the transfer is answered "gone" and the Receiver skips it ("No longer on the phone").

RULES & LIMITS

- The QR code is valid only while its session waits; at most 2 waiting sessions per computer address and 32 in total; sessions expire after 10 min idle.
- Requests after pairing must be signed and come from the same IP: wrong IP → refused; expired session → "expired"; bad signature or replayed counter (already used, or 1024 or more below the highest seen) → refused.
- All answers that carry a decision are signed; a file whose length changed since the list is answered "changed" and re-fetched by the Receiver.

- Resume from a byte offset is supported; data is encrypted in 1 MB frames. The "at most 3 streams" limit is enforced by the Receiver.
- Earlier verifications are matched by the same media id and size; they are labels only and never allow delete for good.

WHAT THE USER SEES

Step 4 when every delivered file is verified: "*N* of *M* files (*size*) were checked on your computer and match exactly. Move them to the trash here." with "Move *N* sent files to trash" and the same dialog as the browser way. If browser and Receiver deliveries are mixed, the browser wording is used.

SAFETY NOTES

The pairing secret travels only through the camera, never over the network. "Verified" = the Receiver's on-disk re-hash equals the phone's whole-file SHA-256 and size.

> **TESTER NOTE**
> Files the Receiver already had are verified without sending bytes, so the phone's progress bar may never reach "All … sent" even though everything is verified. The short code appears only in the Send screen hint, not in the confirmation prompt.

> **DOCS VS CODE**
> The Receiver plan's phone text "Verified on your Windows PC" does not exist; the phone says "…were checked on your computer and match exactly…", and the prompt names the computer's host name, not its system. The protocol says the short code is shown with the Receiver's name before the fingerprint prompt; it is not. The plan says the phone caches file hashes; the code hashes the whole file every time.

Code: `share/Gs2.kt`, `share/Gs2Routes.kt`, `share/FileHasher.kt`, `share/ShareService.kt`

8.4 Send the trash to computer, and delete for good (path D2)

PURPOSE

Copy originals that GallerySlim put in Android's trash to a computer, then optionally delete them for good to free the space now instead of after 30 days.

WHERE

- **All tools** › **Your phone** card → "Send to computer first" (shown when confirmed shrink originals are in the trash).
- Get space back → "Empty what's already done" → "Send to computer first".

HOW IT WORKS

1. Queues: originals of confirmed shrinks, plus clean-ups, duplicates/burst extras, contact-sheet extras and chat media that GallerySlim trashed and that were not undone — each only if still in the trash right now. Sorted by date taken.
2. Earlier Receiver verifications (same id and size) are loaded as labels only. Sharing starts and the Send screen opens.
3. After a send, step 4 shows (first match wins):

- Some verified in this run: "*N* files (*size*) were checked on your computer and match exactly. They're still in the phone's trash; delete them for good to free the space now."
- None now, some earlier: "*N* of these were checked on your computer before. Open the GallerySlim Receiver and send again: it re-checks the files it already has without copying them twice, and then you can delete them for good."
- Browser only: "Files sent through the browser can't be checked, so they stay in the trash, which empties itself after 30 days. To free the space now, send them with the GallerySlim Receiver, which checks every file." (no button)

4. Button "Delete *N* checked files for good" → dialog "Did the files arrive?" / "Only files that were fully sent are included. They'll be deleted from the phone for good, and the copy on your computer will be the only one. Check that they open on your computer first." / "Yes, delete for good" (red), "Not yet".
5. Re-check: the trash is queried again; only files still in the trash **and** verified by the Receiver in this run are kept; then Android's delete dialog opens.

RULES & LIMITS

- Deletes for good only items that GallerySlim trashed, that are still in the trash, and that were verified by the Receiver in the current run ("this run" = since the queue was set; a new queue or stopping sharing resets it). Earlier verifications never qualify. Browser sends never qualify.
- Files trashed by other apps and files already "moved to computer" are not included.

WHAT THE USER SEES

- Nothing qualifies at the start: "Nothing from GallerySlim is in the trash right now" (the Send screen does not open).
- OK: "Deleted *N* files for good. They're on your computer." — recorded in Moved to computer; the ledger refreshes.
- Nothing left at the re-check: "Nothing can be deleted for good — only files verified on your computer in this session" (no Android dialog). Cancel: nothing changes.

SAFETY NOTES

This is one of only two permanent-delete paths. It always needs the in-app dialog and Android's dialog.

> **TESTER NOTE**
> The entry links appear only when shrink originals are in the trash; if the trash holds only clean-up items, there is no way to start this export, although those items would qualify.

Code: `share/TrashExport.kt`, `ui/AppNavHost.kt`, `ui/share/SendToLaptopScreen.kt`

8.5 Moved to computer (archive map)

PURPOSE

Remember what was moved to a computer, with a small thumbnail, its month folder and which computer.

WHERE

All tools › More › Moved to computer.

HOW IT WORKS

- An entry is added only after freeing space from the Send screen (move to trash or delete for good) and Android's dialog returns OK. A 256 × 256 thumbnail is made before the originals leave (deleted again if the dialog is cancelled).
- Screen: header "Moved to computer", summary "*N* items · *size*", one section per month (titled "*yyyy-MM*" or "undated", sorted descending as text, so "undated" first), a line per "*computer* · *folder*", and a 4-column thumbnail grid.

RULES & LIMITS

- Computer label: browser sends — "Windows PC", "Mac", "Chromebook", "Android device", "Linux computer" or "Computer", plus " (*IP*)"; Receiver sends — the Receiver's computer name, or "Your computer".
- Folder label: always "GallerySlim/*yyyy-MM*" (or "GallerySlim/undated").
- All moves ever recorded are shown (no 30-day limit), including ones later undone.

WHAT THE USER SEES

Empty: "Nothing moved yet. Send photos to your computer, then free the space here; this page remembers where each one went."

SAFETY NOTES

Read-only history.

> **DOCS VS CODE**
> The folder label "GallerySlim/*yyyy-MM*" matches the browser ZIP, but the Receiver saves month folders directly inside the chosen folder, and single browser downloads have no folders.

Code: `ui/ArchiveMapScreen.kt`, `history/History.kt`

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9 Other features

9.1 Search

PURPOSE

Find a photo by describing what is in it ("beach", "receipt"), entirely on the phone.

WHERE

Home › **Search**; or All tools → search icon / "Search photos". Header "Search photos"; box hint "Try "beach", "dog", "receipt", "birthday cake"".

HOW IT WORKS

1. While the Search screen is open, photos are indexed (newest first): on-device labels with confidence ≥ 0.6 and Latin-script text (first 600 characters), from a copy at least 960 px on the long edge. Progress: "Getting to know your photos… *N* of *N*. Results grow as this runs; it all stays on your phone."; results refresh every 25 photos.
2. Type a query; results appear as "*N* photos" and a 3-column grid. Tapping a photo opens it in the phone's viewer.

RULES & LIMITS

- Only photos are indexed (no videos). A photo is (re)indexed when new or its size changed. Leaving the screen stops indexing.
- Query: lower-cased, split on non-letters/digits; one-letter words and the words a, an, the, of, in, on, at, my, with, and, photo(s), picture(s), pic, from are dropped. Every remaining word must match a label, a synonym, a label prefix (words of 4+ letters) or appear in the photo's text.
- Synonyms: sea, ocean → beach; kid, kids → child; puppy → dog; meal → food; mountains, hills → mountain; flowers → flower; bill → receipt.
- Ranking: 3 points per label hit, 1 per text hit; then newest first.

WHAT THE USER SEES

Empty query: nothing. No match: "No matches yet." Only photos still in the current scan are shown.

SAFETY NOTES

Read-only; nothing leaves the phone.

> **DOCS VS CODE**
> The round-2 plan describes a background indexer; the code indexes only while the Search screen is open.

Code: `search/PhotoSearch.kt`, `search/SearchViewModel.kt`, `ui/SearchScreen.kt`

9.2 Share small

PURPOSE

Share smaller copies of photos and videos from any app, without changing the originals.

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WHERE

Android's share sheet from any app (photos or videos) → target “Share small”. Not inside GallerySlim's own screens.

HOW IT WORKS

1. Shows a spinner, “Making smaller copies…” (then "… *i* of *n*") and “Your originals stay as they are.”
2. Copies from the previous Share small run are deleted first.
3. Photos: long edge at most 2048 px (never upscaled), turned upright, JPEG quality 82, named `n_name_small.jpg`; metadata, including location, is dropped. Videos: H.264, short edge at most 720 px (never upscaled), about 2.5 Mbit/s, named `n_name_small.mp4`.
4. If a copy is not smaller or fails, a plain copy of the original is shared instead.
5. Android's chooser opens, titled “Share smaller copies” (Share small itself is left out).

RULES & LIMITS

- Up to 50 items; extra items are silently ignored. Only content:// items; if nothing usable is shared, the screen closes at once.
- Names: characters other than A–Z, a–z, 0–9, ".", "_" and "-" become "_"; cut to 80 characters.

WHAT THE USER SEES

If no item could be prepared: “Couldn't read these files.” (no button). A single failed item among several is silently left out.

SAFETY NOTES

Originals are only read. Copies live in the app's cache.

Code: `sharesmall/ShareSmallActivity.kt`, `sharesmall/ShareSmall.kt`

9.3 Safety vault

PURPOSE

Keep an exact copy of every original in a folder the user chooses (ideally an SD card, USB drive or cloud-synced folder) before anything is shrunk.

WHERE

- **Settings** › **Safety** › **Safety vault** (switch): “Keeps an exact copy of every original in a folder you choose, before anything is shrunk”; always-visible note “A vault on this phone's own storage uses as much space as the originals. Pick an SD card, USB drive or a cloud-synced folder.”
- “Open vault” (“See and restore the copies kept in the vault”, or “Choose the vault folder” when none is set) → app lock → screen “Safety vault”.

HOW IT WORKS

- Turning the switch on with no folder opens Android's folder picker; cancelling turns the switch off again. Picking a folder: “Vault location set” or “Couldn't use that folder for the vault”. A subfolder `GallerySlim Vault` is created in the chosen folder.
- Vault screen, no folder: “Keep exact copies, just in case”, explanation, button “Choose vault folder”.

- Folder set: header action "Change folder"; if the folder is on the phone's internal storage, a warning "This vault folder is on the phone's own storage, so every copy uses as much space as the original and nothing is freed. Tap Change folder and pick an SD card, a USB drive or a cloud-synced folder."
- Empty: "The vault is ready. Originals are copied here automatically before each shrink while the vault is enabled in settings."
- With entries: "*N* originals protected · *size*"; rows with name, "*size* · *date archived*" and a red "Restore".

RULES & LIMITS

- Copies are made only when the vault is on **and** a folder is set, before every photo/video shrink, light copy, trim and motion strip (also overnight and trips) — so items later skipped also get a vault copy.
- Exact byte copies (no JPEG XL). Each copy is read back from the vault folder and accepted only if its length and SHA-256 match the original; otherwise the partial copy is deleted and the item is not shrunk ("Couldn't protect it in the vault — left untouched").
- Vault on but no folder: manual batches are refused ("Pick a vault folder first"); overnight/trip runs are skipped with a notification.
- Vault names: / \ : * ? " < > | and control characters become "_"; leading/trailing spaces and trailing dots removed; at most 127 characters (extension kept); empty → "unnamed"; clashes (ignoring case) → "IMG_001 (2).jpg", "(3)", …
- Restore (no confirmation): copies the original back into the gallery under its original name (hidden until complete; read back and checked for the same length and SHA-256 as the vault copy, otherwise removed). The vault file stays; the entry leaves the list. The shrunk copy is not touched, so both files are then in the gallery. "Change folder" does not move existing entries.

WHAT THE USER SEES

Restore results: "Restored *name* to your gallery" or "Couldn't restore *name*".

SAFETY NOTES

The vault folder survives uninstalling the app. Off by default.

> **DOCS VS CODE**
> The original design describes a JPEG XL vault and suggests the phone's Documents folder; the code stores exact copies and warns against phone storage.

Code: `vault/VaultManager.kt`, `vault/VaultNaming.kt`, `ui/VaultScreen.kt`

9.4 App lock

PURPOSE

An optional fingerprint/face/PIN lock in front of the originals waiting to be cleared and the vault.

WHERE

Settings › **Safety** › **Lock with fingerprint or PIN**, caption "Asks for your fingerprint or PIN before showing the originals that are ready to clear, and the vault". Off by default.

HOW IT WORKS

- Prompt "Unlock GallerySlim" / "Confirm it's you to see what's ready to clear and the vault" (biometric or the phone's PIN, pattern or password).
- Asked on: Home Ready-to-clear card; All tools Ready-to-clear banner; Get space back "Review and clear"; progress screen "Go to Ready to clear" and Compare links; Settings "Open vault"; and notification taps that open Ready to clear, the vault or Recent changes.
- Once unlocked, the app stays unlocked until its process ends. Cancel/failure: nothing happens, no message.

RULES & LIMITS

Turning the lock on with no screen lock or biometrics set up on the phone: "Set a screen lock or fingerprint on your phone first" and the switch stays off. Turning it on does not ask for authentication.

WHAT THE USER SEES

Android's system prompt.

SAFETY NOTES

Separate from the app lock, Send to computer always asks for confirmation — the fingerprint/PIN prompt, or an in-app dialog on phones without a screen lock (see Send to computer).

Code: `ui/AppLock.kt`, `ui/AppNavHost.kt`

9.5 Notifications

PURPOSE

Progress and results of background work, and gentle reminders.

WHERE

Android's notification shade. Channels can be managed in Android's app settings.

HOW IT WORKS

Channel (importance)	Title / text	When	Tap opens
Overnight shrink progress (low)	"Shrinking your gallery" / "*done* of *total* · *name*" or "Scanning…" (ongoing)	Manual batch, overnight run, trip	Nothing
Overnight shrink progress (low)	"Ready to send to your computer" / "Open the link on the computer and scan its code · stops by itself after 30 min", action "Stop sharing"	While sharing runs	The app
Overnight shrink results (default)	"Overnight shrink finished" / "Ready to clear: *N* originals · *size*. Tap to clear them.", or "Overnight shrink — nothing to do" / "Your gallery is already slim."	After an overnight run (no storage goal)	Ready to clear
Overnight shrink results	"Below your free-space goal" or "Overnight shrink finished" / goal text (see Storage floor)	After an overnight run with a goal	Ready to clear

Channel (importance)	Title / text	When	Tap opens
Overnight shrink results	"Overnight shrink skipped" / "The vault is on but no vault folder is set. Pick one to resume overnight shrinking."	Overnight or trip run, vault without folder	Home
Overnight shrink results	"Ready for your trip", "Trip prep: *N* items shrunk" or "Trip prep finished" (texts in Trip mode)	Trip job	Ready to clear
Overnight shrink results	"Last chance to restore *N* originals" / "They leave your phone's trash on *d MMM*. Open your gallery's trash to restore any you want back."	Daily; confirmed shrink originals leaving the trash within 3 days; once per expiry day	Nothing
Monthly report (low)	"Your *Month Year*: *X* freed" / "Room for about *N* more photos. Tap to see and share."	First daily check of a new month; skipped if nothing was freed	Your month
Weekly tidy-up (low)	"Your weekly tidy-up is ready" / "Your rules picked what to clear. Take a look, then confirm once."	Every 7 days if rule 1, 2 or 3 is on	Quick review
Low space (default)	"Running low: *X* GB free" / "Make room now, before the camera says it's full?"	Below 2 GB free, at most once per 24 h	Make room

RULES & LIMITS

- If notifications are denied, all work still runs silently; Home and All tools still show what is waiting.
- The morning note and the goal note replace each other (same notification slot).
- When Android's daily time limit for background work stops a batch, overnight run or trip: "Android's daily time limit for background work was reached. What finished is in Ready to clear; the rest continues next time." (trip title "Trip prep paused").
- Taps that open Ready to clear go through the app lock (if on).
- The trash-expiry reminder's expanded text: "*N* originals GallerySlim replaced with smaller copies leave your phone's trash on *date*. Open your gallery's trash (Recycle bin) to restore any you want back."

WHAT THE USER SEES

As in the table.

SAFETY NOTES

No notification action deletes anything; the only action button is "Stop sharing".

> **TESTER NOTE**
> The trip result notification uses the same slot as the manual-batch progress notification, so a later manual batch may replace it. The sharing notification sits under the channel named "Overnight shrink progress".

Code: `work/ShrinkNotifications.kt`, `work/TrashWatchWorker.kt`, `report/MonthlyReport.kt`, `cleanup/StandingRules.kt`, `storage/StorageWidget.kt`, `share/ShareService.kt`

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9.6 Settings (every option)

PURPOSE

All user settings in one bottom sheet.

WHERE

The tune icon on Home or on All tools. Title "Settings"; section titles in capitals.

HOW IT WORKS

Section	Option	Choices (default)	Effect
"Quality"	"How much to shrink"	Gentle (default), Balanced, Aggressive, Light copy	Global preset; see Presets. Caption on GPS loss.
"Automatic"	"Shrink overnight"	On / off (off)	Schedules the overnight job; asks for notification permission when turned on
"Automatic"	"Keep this much space free"	Off (default), 5, 10, 15, 20 GB	Storage floor
"Automatic"	"Lighter copies of old photos"	Off (default), 1, 2, 3 years	Cooling
"Automatic"	"Your rules"	—	Opens the five questions
"Safety"	"Protect new photos for"	7, 30 (default), 90 days	Protect window
"Safety"	"Lock with fingerprint or PIN"	On / off (off)	App lock
"Safety"	"Safety vault"	On / off (off)	Vault; picker opens when turned on without a folder
"Safety"	"Open vault"	—	App lock → Safety vault screen
"Other"	"Motion-clip export folder"	Folder (not set)	Where clips are exported before removal
"Other"	"Privacy"	—	Opens the Privacy page ("What GallerySlim does and doesn't do with your photos")

RULES & LIMITS

- Answering standing-rule questions 4 and 5 immediately overwrites the storage goal and cooling here.
- Not in Settings: language, theme, app version and the diagnostic log.
- App data (settings, history, search index) is excluded from Google backups and phone-to-phone transfers, so a restored or new phone starts fresh.
- The app never displays its version. The app follows the phone's light/dark mode.

WHAT THE USER SEES

Captions under each option, as quoted in the individual feature sections.

SAFETY NOTES

No setting deletes anything.

Code: `ui/SettingsSheet.kt`, `res/values/strings.xml`

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9.7 Privacy page and "How it works" page

PURPOSE

Explain in plain words what the app does with photos, and how shrinking stays safe.

WHERE

Settings › **Privacy**; **All tools** › **More** › **How it works**.

HOW IT WORKS

Privacy (header "Privacy"), six sections:

1. "Everything happens on your phone" — "Photos and videos are shrunk, checked and sorted on this phone. They are never uploaded to us or anyone else."
2. "No account, no ads, no tracking" — "GallerySlim has no sign-in, shows no ads and contains no analytics or tracking code."
3. "Internet is used only for Send to computer" — "The app uses the network only to send photos to a computer on your own Wi-Fi. Sharing runs while its notification is showing (tap Stop sharing to end it), sends only to the computer whose code you scanned and confirmed with your fingerprint or PIN (or, on a phone without a screen lock, with a confirmation message), and stops by itself 30 minutes after the last transfer or when Wi-Fi drops. Nothing goes over the internet."
4. "Nothing is deleted without you" — "Originals move to Android's trash only after you confirm, and stay restorable there for 30 days. Two things delete for good, each only after you confirm in the app and again in Android's own dialog: "Free this space now" (originals that already have a checked smaller copy) and "Send the trash to computer" (files the GallerySlim Receiver has checked byte for byte on your computer)."
5. "Text reading stays on the phone" — "Clean up reads text in photos to spot receipts and ID documents. This uses an on-device model; no image or text leaves the phone. ID documents are never suggested for deletion."
6. "Your diagnostic log stays here" — "The app keeps a small log of what it did, on the phone. It is shared only if you tap Share diagnostic log yourself. App data is excluded from Google backups and phone-to-phone transfers."

How it works: three numbered steps — "GallerySlim makes a smaller copy", "The original waits in Ready to clear", "You clear the originals" — and "Common questions": "Will my photos look worse?", "Can I undo?", "Does anything get uploaded?", "What about Google Photos?", "Which photos are left alone?" (answer: "Favourites, photos from the last 30 days, HDR photos, ID documents and albums you protect. You can still pick them by hand.").

RULES & LIMITS

Static text.

WHAT THE USER SEES

As above.

SAFETY NOTES

These pages make promises a tester should check against behaviour; the remaining gaps are below. `docs/privacy-policy.md` carries the same promises in longer form (plus Permissions and Contact sections not shown in the app).

DOCS VS CODE

- "Stops by itself 30 minutes after the last transfer": in code sharing stops about 30 minutes after the last approved session expires, and sessions expire after 10 idle minutes, so about 40 minutes after the last transfer.
- "Share diagnostic log" is only on the progress screen's skip card, not in Settings.
- "Photos from the last 30 days" is the adjustable protect window (7/30/90). HDR photos are not fully left alone: Ultra HDR JPEGs get a lossless re-optimisation.

Code: `ui/PrivacyScreen.kt`, `ui/HowItWorksScreen.kt`

9.8 Hindi

PURPOSE

Show the Settings sheet in Hindi for Hindi-language phones.

WHERE

Automatic when the phone's system language is Hindi. There is no in-app language option.

HOW IT WORKS

Only the 44 text resources are translated: the Settings sheet (title, sections, every option and caption, preset names, chip labels) and the widget's description in the widget picker. "GallerySlim" is not translated. Every other screen, dialog, snackbar and notification, the widget's own labels and the "Share small" label are English.

RULES & LIMITS

English	Hindi (as shipped)
Settings	सेटिंग्स
How much to shrink	कितना छोटा करें
Gentle / Balanced / Aggressive / Light copy	हल्का / संतुलित / ज़्यादा / हल्की कॉपी
Quality / Automatic / Safety / Other	क्वालिटी / अपने आप / सुरक्षा / अन्य
Shrink overnight	रात में छोटा करें
Keep this much space free	इतनी जगह हमेशा खाली रखें
Lighter copies of old photos	पुरानी फ़ोटो की हल्की कॉपी
Your rules	आपके नियम
Protect new photos for · *N* days	नई फ़ोटो को सुरक्षित रखें · `%1$d` दिन
Lock with fingerprint or PIN	फ़िंगरप्रिंट या PIN से लॉक करें
Safety vault · Open vault	सेफ़्टी वॉल्ट · वॉल्ट खोलें
Motion-clip export folder · Not set	मोशन-क्लिप एक्सपोर्ट फ़ोल्डर · सेट नहीं है
Privacy	प्राइवेसी
Off · *N* GB · Older than *N* yr	बंद · `%1$d` GB · `%1$d` साल से पुरानी

In the table, %1$d stands for the number the app fills in.

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WHAT THE USER SEES

Sizes always use Western digits and a dot decimal ("4.2 GB").

SAFETY NOTES

None specific.

> **DOCS VS CODE**
> Design v2 lists "Hindi strings" as a feature; in code it covers only Settings. Clean up's greeting detection reads romanised Hindi but not Devanagari script.

Code: `res/values-hi/strings.xml`, `ui/SettingsSheet.kt`

9.9 Diagnostic log

PURPOSE

A small on-phone log that the user can choose to share when reporting a problem.

WHERE

Progress screen → card "Why some weren't shrunk" (shown only when something was skipped or failed) → "Share diagnostic log".

HOW IT WORKS

Opens Android's share sheet ("Share diagnostic log", subject "GallerySlim diagnostic log") with the current log file. If the log is empty or missing, nothing happens.

RULES & LIMITS

- Stored in app-private storage; rotates above 1 MB to one backup (only the current file is shared).
- Contains names, sizes, flags, decisions, crash traces and the device model/Android version; never file contents or EXIF/GPS values.

WHAT THE USER SEES

Android's share sheet.

SAFETY NOTES

Shared only when the user taps the button.

Code: `diag/GsLog.kt`, `diag/DiagShare.kt`, `ui/ShrinkProgressScreen.kt`

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10 GallerySlim Receiver (laptop program)

A single program for Windows, macOS and Linux (version 0.1.0) that copies photos from the phone over Wi-Fi into one folder the user chooses, sorted into month folders. It only connects *to* the phone; it opens no network port. Its own page runs in the computer's browser on 127.0.0.1. A file counts as safe only after the Receiver has read it back from its own disk and the phone has confirmed the bytes match ("verified").

10.1 Which file, and first run on each system

PURPOSE

Start the Receiver on any of the three systems without installing anything else.

WHERE

Computer	File
Windows 10/11 (most PCs)	`windows-amd64/GallerySlim-Receiver.exe`
Windows on ARM	`windows-arm64/GallerySlim-Receiver.exe`
Mac with Apple silicon	`GallerySlim-Receiver-macos-arm64.zip`
Mac with Intel	`GallerySlim-Receiver-macos-amd64.zip`
Linux PC	`GallerySlim-Receiver-linux-amd64.tar.gz`
Linux on ARM (64-bit)	`GallerySlim-Receiver-linux-arm64.tar.gz`

`SHA256SUMS` lists every file's checksum. Minimum systems: Windows 10, macOS 13, any recent 64-bit Linux.

HOW IT WORKS

- **Windows**: double-click the .exe. A console window opens next to the browser page; that window is the Receiver — close it to stop.
- **macOS**: unzip, move "GallerySlim Receiver" to Applications, open it. No Dock icon; it shows its page in the browser. Allow "Local Network" access when asked (without it the phone cannot be seen). Quit with "Close the receiver" on the page.
- **Linux**: unpack the .tar.gz and run `./GallerySlim-Receiver`. Without a desktop it runs in the terminal (see Headless mode).

RULES & LIMITS

- Builds are unsigned (signing pending company IT). Per the README: Windows SmartScreen needs **More info → Run anyway**; Windows 11 Smart App Control blocks unsigned programs with no override; macOS 13–14 needs right-click → Open; macOS 15+ needs System Settings → Privacy & Security → Open Anyway; a "damaged" message means the download is quarantined.
- Windows Firewall may ask about network access; allow private networks.

WHAT THE USER SEES

Console (page mode): "GallerySlim Receiver *0.1.0* is running.", "If your browser didn't open, go to:" and the address, then "Keep this window open while receiving. Close it or press Ctrl+C to stop." If

the browser cannot be opened: “Couldn't open the browser (*error*); open the address above yourself.”

SAFETY NOTES

The OS warnings for unsigned builds are expected in this test phase; they are not defects of the Receiver.

Source: `receiver/README.md`, `receiver/build.sh`, `receiver/main.go`

10.2 Choosing the folder, and the cloud warning

PURPOSE

Photos go into one folder the user chooses; the Receiver steers away from folders that sync to the cloud, because synced photos may be moved off the computer and can then no longer be checked.

WHERE

The page's first-run phase (shown on first run, or when the saved folder is no longer writable), or `--folder` on the command line.

HOW IT WORKS

- Heading “Where should your photos go?”, text “Your photos will be saved in this folder, sorted into one folder for each month.”, label “Suggested folder” and the path (or “No folder suggested. Choose one, please.”).
- “Choose folder” opens the system's folder dialog titled “Choose where to save photos”; message “A folder window has opened. It may be behind this browser window.” On success: “Saved. Photos will go to *path*”. Cancelling makes no change.
- “Use this folder” (when a folder is suggested).
- “Type a folder path” → field “Folder path” and “Use this path” (Enter works). Quotes from Windows "Copy as path" are stripped; a path may start with `~` (the home folder), which is expanded.
- The folder is tested by creating `.gallery-slim/` and a test file in it.

RULES & LIMITS

- **Suggestion**: if the Pictures folder does not sync, `<Pictures>/GallerySlim Photos`; if Pictures syncs, `<home>/GallerySlim Photos` (with a note); if both sync, Pictures itself with a warning.
- **Dialogs**: Windows' own folder dialog; macOS "choose folder"; Linux zenity or kdialog (kdialog first on KDE). Without a dialog program: “This computer can't show a folder picker; type the folder path.” and the text box opens.
- **Cloud detection** (folder names anywhere in the path; case-insensitive on Windows/macOS): `OneDrive` or "OneDrive - …"; `Google Drive` or `My Drive`; `Dropbox` or "Dropbox (…"; `iCloudDrive` / `iCloud Drive`; `iCloud Photos`. Windows: the OneDrive folders from the system. macOS: `~/Library/Mobile Documents` (iCloud Drive) and `~/Library/CloudStorage/…`. Linux: `~/Insync`, Google Drive and OneDrive network mounts. Other services (pCloud, Mega, Nextcloud…) are not detected.
- Path checks: empty → “Please choose a folder, or type its full path.”; over 4096 bytes → “That folder path is too long.”; control/bidi characters → “That folder path has characters that can't be

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used."; not absolute → "Please type the full folder path, for example D:\Photos" (macOS/Linux examples differ).
- Not writable: "Photos can't be saved in that folder: *cause*". During receiving: "Please wait until receiving has finished".
- Once chosen, the page has no control to change the folder; use `--folder` or edit `config.json`.

WHAT THE USER SEES

Cloud warning (not blocking): "This folder syncs to *provider*. Photos there can be moved to the cloud and removed from this computer, so they can't be checked here. A folder that stays on this computer is safer." Suggestion note: "Your Pictures folder syncs with *provider*, which can move photos to the cloud and leave only placeholders on this computer." …

SAFETY NOTES

A synced folder can make "verified" untrue later (files replaced by placeholders); the warning exists for that reason.

> **DOCS VS CODE**
> The plan suggests "the real Pictures location"; the code suggests a "GallerySlim Photos" subfolder. The README says the suggestion is never synced; when both Pictures and home sync, the code suggests Pictures with a warning.

Code: `receiver/internal/desktop/*.go`, `internal/app/app.go`, `internal/ui/*`

10.3 The Receiver page: phase by phase

PURPOSE

Guide the user from folder choice to "All done", with plain status at every step.

WHERE

The browser page at `http://127.0.0.1:port/token/`. Header "GallerySlim Receiver"; footer "Version *v*" and "Close the receiver" (hidden while receiving).

HOW IT WORKS

Phase	What is shown
Starting	"Starting…" with a spinner
Choose folder	See the previous section
Finding	"Looking for your phone…", "On the phone, open **GallerySlim** → **Send to computer**." Searches again every 8 s. Not found: "No phone found yet. On your phone, open GallerySlim and tap "Send to computer". Both need to be on the same Wi-Fi. You can also type the address shown on the phone." Several: "Phones found. Choose yours:" with buttons "*name* · code *code*" (re-search stops until one is picked). One phone: goes straight to pairing. Button "Search again". Card "Enter the phone's address": hint "It is shown on the phone's Send to computer screen, for example 192.168.1.23:8765.", button "Connect". Footer "Photos will be saved to *folder*".
Pairing	"Scan this code with your phone", the QR code, "Code *K7-42*", "Phone: *name*", "On your phone, tap **Scan** and point it at this code. Check the phone shows ***code***." and "Saving to *folder*". No buttons except Close the receiver.

Phase	What is shown
Receiving	"Receiving your photos", a progress bar, "*done* of *N* photos · *x* of *y*", "Now: *file*", "Saving to *folder*", "Keep this window open until it says Done.", and "Recent files" (newest first, up to 500 rows, then "and *N* earlier file(s)"). Tab title "*NN*% · GallerySlim Receiver".
Done	"All done!" or "Finished, but some photos were not saved"; summary lines (below); "Your photos are in *folder*"; buttons "Open folder" and "Receive more"; a "Not saved" list. Tab title "Done · GallerySlim Receiver".
Error	"Something went wrong", the error (or "The transfer stopped."), "Check that the phone and this computer are on the same Wi-Fi and GallerySlim is open on the phone.", "Photos already saved on this computer are kept.", button "Try again".
Closed	"The receiver is closed" / "You can close this browser tab now." The program exits about 150 ms later.

File row states: ○ "Waiting" · ↓ "Receiving…" · ✓ "Saved, checking…" · ✓ "Checked and safe" · – "Skipped" · ! "Not saved".

Done summary: "*n* photo(s) saved and checked. The phone knows they are safe here." · "… saved but not confirmed by the phone. Keep them on the phone for now." · "… skipped." · "… could not be saved. They are still on your phone, so you can try again with Receive more."

RULES & LIMITS

- The page polls every 700 ms (2 s after failures). After 3 failed polls it shows "Lost contact with the receiver." If you closed it, open GallerySlim Receiver again.
- "Receive more": if some files failed, retries only the missing files in the same session (already verified ones show "Received and checked earlier"); otherwise searches for a phone again.
- "Try again" after a pairing timeout makes a new code for the same phone; after other errors it searches again.
- Code expired: "The code expired before it was confirmed on the phone. Press Try again for a new code." Phone unreachable: "Couldn't reach the phone. Check that it still shows "Send to computer" and is on the same Wi-Fi."
- General errors: "Could not reach the receiver. Is it still open?", "That didn't work. Please try again.", "Please wait a moment, the receiver is still working on your last request.", "This isn't available right now.", "Something went wrong inside the receiver. Please try again."

WHAT THE USER SEES

"Open folder" opens Explorer, Finder or the Linux file manager ("No folder chosen yet" if none).

SAFETY NOTES

Phone-supplied text (name, code, file names) is cleaned of control and direction-changing characters and length-limited before display.

> **TESTER NOTE**
> "Saved, checking…" is also the final label for files that were saved but not confirmed (only the detail text says so). If the phone ends the session, the page returns to pairing with a new code; its explanatory note is set but not shown.

Code: `receiver/internal/ui/page.js`, `page.go`, `ui.go`, `internal/app/app.go`

10.4 Finding the phone

PURPOSE

Find the phone that is showing "Send to computer" on the same Wi-Fi, automatically or by typed address.

WHERE

The Finding phase; or `--phone`.

HOW IT WORKS

1. One multicast name query for `galleryslim.local`; answers collected for 1.5 s; then the phone's ports 8765–8769 are tried at those addresses.
2. If nothing answers: every address of each active IPv4 network with a /24 to /30 prefix is probed on 8765–8769 (300 ms timeout, 64 in parallel), excluding network, broadcast and the computer's own address.
3. A phone counts only if it answers as GallerySlim protocol 2.

RULES & LIMITS

- Typed address: IPv4 with optional port, at most 64 characters (page: at most 100 before cleanup; http:// and paths are stripped). Host names such as galleryslim.local are rejected. Without a port, 8765–8769 are tried.
- Page messages: “Please type the address shown on the phone.”, “That address is too long.”, “That doesn't look like the phone's address. It looks like 192.168.1.23:8765.”; no answer: “no GallerySlim phone answered at *addr*; check that the phone shows "Send to computer" and is on the same Wi-Fi”.
- Networks wider than /24 (e.g. /16) are not probed: type the address.

WHAT THE USER SEES

The Finding phase texts above.

SAFETY NOTES

Discovery only reads the phone's public hello; nothing is sent until pairing is approved on the phone.

> **DOCS VS CODE**
> README/protocol say "/24 or smaller"; the code also skips /31 and /32.

Code: `receiver/internal/discover/*.go`, `internal/client/client.go`

10.5 Pairing, transfer, resume and verification

PURPOSE

Pair securely through the camera, download every file, and prove each saved file matches the phone's.

WHERE

Pairing and Receiving phases (page or terminal).

HOW IT WORKS

1. The Receiver makes a 32-byte secret, shows it only inside the QR code (`gs2:…`) with the phone's short code, and polls the phone every 1 s (5 s timeout per poll).
2. After the owner confirms on the phone, the file list is fetched (up to 3 tries, waits 1 s and 2 s).
3. Free-space check: sizes of files not already present, minus partial downloads, plus a 50 MB margin. Short → error "not enough free space: need *X*, free *Y*" (the "need" figure excludes the margin).
4. Sizes on the page and in the terminal use decimal units (1 kB = 1000 bytes) with one decimal below 10 and whole numbers from 10 up, e.g. "4.2 GB", "245 MB" (the terminal writes kilobytes as "kB"; below 1000 bytes it shows bytes).
5. Up to 3 files download in parallel, encrypted. Each file is written as a partial file, moved into place without overwriting, **re-read from disk and hashed**, and the hash and size are sent to the phone, which answers "verified" or not.

RULES & LIMITS

Situation	Behaviour
Pairing	Times out after 10 min, or after 15 failed polls in a row
Phone busy (503)	Wait 2 s, retry up to 5 times, then "The phone stayed busy"
No data for 60 s	Stream cancelled ("The connection stopped; try again") and retried
Per-file retries	5 attempts, waits 1, 2, 4, 8 s (max 15 s)
Resume	Partial files in `<folder>/.gallery-slim/parts/`, named from phone, file id, size, date and hash, so a new QR session with the same phone continues them. A partial file larger than the listed size is discarded
Partial-file cleanup	Untouched for more than 14 days → deleted at the start of each run
File changed on the phone	Partial discarded, list fetched again (at most 2 times); gone from the phone → "Skipped" "No longer on the phone"
Checksum differs from the phone's list	Downloaded once more from scratch; then "Not saved" "The received copy doesn't match the phone's checksum"
Wrong size on disk after saving	Failed: "The saved file has the wrong size on disk"
Confirmation could not be sent	Saved but not confirmed: "Saved, but the phone could not be asked to confirm it"
Phone says the copy does not match	File kept and flagged; one fresh download saved under a new name; if still not confirmed: "Couldn't confirm — kept. The phone says this copy doesn't match, so it won't be offered for freeing space."
File already on the computer	Re-hashed on disk and confirmed without downloading: "Already on this computer (checked)"
Windows path ≥ 260 characters	Saved and verified, with the note "The full path is longer than 260 characters; some Windows programs can't open it"
List limits	List ≤ 64 MB, ≤ 200,000 files, file ≤ 1 TB, id ≤ 256 bytes; invalid or duplicate entries silently dropped

WHAT THE USER SEES

Row states and summary on the page; one line per file in the terminal. Short errors: "The phone ended the session", "The phone did not accept this computer's requests", "The phone refused this computer's address", "The phone stayed busy", "The connection stopped; try again", "Stopped".

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SAFETY NOTES

- The secret is never sent over the network, written to disk or logged.
- Answers from the phone are signed; an answer without a valid signature is treated as a network error, never as a decision.
- A finished file is never deleted or replaced because of anything the phone says.

> **DOCS VS CODE**
> The protocol limits ids to 64 characters; the Receiver accepts up to 256 bytes. The plan's single "free-space check" also needs a 50 MB margin in code.

Code: `receiver/internal/receive/*.go`, `internal/client/*.go`

10.6 Where files go and never-overwrite naming

PURPOSE

Save every photo into a month folder, with a name that works on every system, without ever overwriting an existing file.

WHERE

The chosen folder.

HOW IT WORKS

- Layout: `<folder>/yyyy-MM/name`, or `<folder>/undated/name` when the month is unknown or malformed. Month folders sit directly inside the chosen folder.
- The file's modification time is set to the date taken (unless it is ≤ 0 or more than 24 h in the future).
- Hidden folder `<folder>/.gallery-slim/`: `parts/` and `manifest.json` (a hint only; files are always re-checked).

RULES & LIMITS

- **Never overwrite**: tries `name`, `name (2).ext` … up to `(9999)`; each name reserved with create-exclusive; on Windows and macOS names differing only in letter case count as taken; one lock per folder; file and folder synced to disk; renames briefly locked by antivirus or sync clients retried 6 times.
- **Name cleaning** (all systems): control and direction characters dropped; `<>:"/\|?*` → `_`; trailing dots/spaces and leading spaces trimmed; a leading "." gets a "_" prefix; empty → `photo`; Windows device names get a "_" prefix (e.g. `CON.jpg` → `_CON.jpg`; CON, PRN, AUX, NUL, CONIN$, CONOUT$, COM0–9, LPT0–9 and superscript variants); a missing extension is added from the file type; at most 255 bytes.

WHAT THE USER SEES

Files appear in month folders; a second file with the same name becomes "IMG_1 (2).jpg".

SAFETY NOTES

No existing file is ever replaced, including files from another phone with the same names.

Code: `receiver/internal/store/*.go`

10.7 Keep-awake and idle auto-quit

PURPOSE

Stop the computer sleeping mid-transfer, and stop the Receiver when it is no longer used.

WHERE

Automatic.

HOW IT WORKS

- Keep-awake from after the free-space check until the run ends: Windows keeps the system awake (the display may still turn off); macOS uses `caffeinate`; Linux uses `systemd-inhibit` when available (nothing otherwise). Released if the Receiver dies.
- Idle quit (page mode, all systems): when the page has not been open (polling) for more than 30 minutes and nothing is being received, the Receiver stops (checked every minute, so 30–31 min). Console: "The page has been closed for a while; stopping." Waiting for pairing counts as idle.

RULES & LIMITS

30 minutes; check every 1 minute.

WHAT THE USER SEES

The computer does not sleep while receiving. If the Wi-Fi drops or the laptop sleeps anyway: open the Receiver again and scan again; finished files are kept and partial files continue.

SAFETY NOTES

Stopping never deletes saved files.

> **DOCS VS CODE**
> The README describes the 30-minute auto-quit under macOS only; it applies on every system in page mode.

Code: `receiver/main.go`, `internal/desktop/*.go`

10.8 Headless mode, command-line options and exit codes

PURPOSE

Run the Receiver in a terminal only (Linux without a desktop, or by choice), for example on a home server.

WHERE

`./GallerySlim-Receiver --headless --folder ~/Pictures/Phone [--phone 192.168.1.23:8765]`. On Linux/BSD it switches to headless automatically when both `DISPLAY` and `WAYLAND_DISPLAY` are empty; Windows and macOS never do.

HOW IT WORKS

Option	Meaning
`--headless`	"run in the terminal only and print the code as text"

Option	Meaning
`--folder <path>`	"folder to save photos in" (also remembered in page mode); `~` expanded
`--phone <ip[:port]>`	"the phone's address as shown on the phone, like 192.168.1.23:8765"
`--no-browser`	Don't open the browser; the page address is printed
`--plain-qr`	Terminal QR without colours (default on Windows)
`--version`	Prints the version and exits with 0

Terminal output in order: "GallerySlim Receiver *v*, saving to *folder*" · "Looking for your phone (open GallerySlim and tap "Send to computer")…" · "Found *name* (code *code*) at *addr*" · "Scan this with GallerySlim on your phone. Check that the phone shows the code *code*." + the text QR · "Waiting for you to confirm on the phone…" · "Receiving *n* files (*size*)…" · progress every 5 s "*x* of *y* (*p*%)" · one line per file "[*n*/*total*] verified|saved|skipped|failed *month/name* — *detail*" · "Done: *N* verified (*N* were already here), *N* saved but not confirmed, *N* no longer on the phone, *N* failed."

RULES & LIMITS

Exit	Meaning
0	Nothing failed and nothing is "saved but not confirmed" (files no longer on the phone are allowed). Page mode: closing normally, Ctrl+C or idle quit.
1	Any failed or unconfirmed file; "No phone found. Run again with --phone <address shown on the phone>."; "Pairing failed: …"; "Stopped: …"; "Can't use the folder: …"; a `--phone` address that does not answer or is malformed; Ctrl+C in headless mode; page could not start.
2	Bad command line (unknown flag, stray argument); missing `--folder`; folder not writable ("Photos can't be saved in *folder*: …"); several phones found ("Found several phones; run again with --phone and one of these addresses:" plus one line per phone).

- Explicit `--headless` always needs `--folder` ("Please choose a folder: --headless --folder <path> [--phone <ip[:port]>]"), even when a folder is saved. Automatic headless uses the saved folder; with none: "No desktop found, so running in the terminal. Choose a folder with --folder <path>." (exit 2).
- If the phone ends the session, a new code is shown ("The phone ended the session (…). Scan the new code to continue; saved files are kept."); after 3 sessions: "Gave up after 3 sessions."
- Synced folder note: "Note: *folder* syncs to *provider*; photos there may be moved off this computer."

WHAT THE USER SEES

The QR code is drawn with half-block characters (black on white by default; plain with `--plain-qr`, assuming light text on a dark background).

SAFETY NOTES

Same verification and never-overwrite rules as page mode.

> **DOCS VS CODE**
> The README says exit code 2 covers "several phones found"; a malformed or unreachable `--phone` exits 1, not 2.

Code: `receiver/main.go`, `internal/app/headless.go`, `internal/qr/render.go`

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10.9 Limits, settings, log and page security

PURPOSE

What a tester can check about the Receiver's own files and its local page.

WHERE

The user settings folder and the local page.

HOW IT WORKS

- Settings folder `GallerySlim Receiver` inside `%AppData%` (Windows), `~/Library/Application Support` (macOS) or `~/.config` (Linux), private to the user.
- `config.json`: the folder (absolute only) and `openBrowser` (default true; no UI for it). Written atomically.
- `receiver.log`: private; rotates at 2 MB to `receiver.log.1`; never contains the secret, QR text, keys, signatures or the page token.

RULES & LIMITS

Request to the local page	Result
Host other than 127.0.0.1:*port* or localhost:*port*	403 "Forbidden"
Wrong or missing token, unknown path	404
Wrong method	405
Action from another origin	403
Action not sent as JSON	415
Body over 16 KB	413
Unknown JSON fields	400

- The page listens on 127.0.0.1 only, on a random port, with a random 32-character token in the address. Strict content-security policy; no cross-origin access; cannot be framed.
- Text from the phone is limited: phone name 40 characters, code 12, file name 200, detail 300, errors 400; at most 10 phones and 500 rows shown.

WHAT THE USER SEES

Nothing, unless something is wrong.

SAFETY NOTES

Other programs and web pages on the computer cannot drive the Receiver's page.

Code: `receiver/internal/ui/ui.go`, `page.go`, `internal/app/config.go`, `internal/applog/applog.go`, `internal/textsafe/textsafe.go`

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Appendix A Where the documents and the code disagree

The design documents (`docs/2026-10-17-galleryslim-design.md` "original design", `docs/GallerySlim-Design-v2.md`, `docs/design/ui-spec.md`, the plans in `docs/plans/`, `docs/protocol-gs2.md`, `docs/privacy-policy.md` and `receiver/README.md`) were compared with the code. In every case this reference describes the code. Each item says what the code does.

A.1 Shrinking and quality

Document says	Code does
Original design §4: Gentle = Jpegli JPEG→JPEG at SSIMULACRA2 ≥ 90; Balanced HEIC ≥ 87; Aggressive HEIC ≥ 75	All presets make HEIC photo copies (lossless JPEG fallback only). Floors: SSIMULACRA2 80 / 72 / 60 / 55 (Light copy); SSIM 0.985 / 0.97 / 0.93 / 0.90. No Jpegli.
Original design §6 rule 3: date, GPS and orientation copied to every output	HEIC copies carry no EXIF: orientation baked into pixels, date kept in the gallery's date-taken field, GPS lost. Lossless JPEG copies keep all EXIF. (Design v2 §14 and Settings state the GPS loss.)
Original design §6 rule 2: Ultra HDR skip-only in v1	Ultra HDR JPEGs get a lossless re-optimisation (pixels and gain map kept) when it saves ≥ 1.5 %; otherwise skipped.
Original design §6 rule 7: last 30 days excluded	Adjustable protect window 7 / 30 / 90 days, default 30; protected items can be ticked by hand.
Original design §5: VMAF for video; AV1 where hardware exists	5-frame SSIM spot check (each ≥ 0.90) plus banding guard; output always HEVC.
Original design Phase 5: JPEG XL vault; Documents folder suggested	Exact byte copies; the app warns against the phone's own storage.
Original design §6 rule 1: permanent delete in exactly one place	Two places (D1 Free this space now; D2 Send the trash to computer), matching the 2026-10-22/24 addenda and Design v2.
Round-2 plan: trimmed videos have no Compare	The Compare button is shown on trimmed-video rows; it opens "Can't compare this item".
Round-2 plan: "Light copy (720p)"	Button reads "Make 720p light copies".
Round-2 plan: album rules for overnight and manual shrink	Also applied to Make room, trips, Big videos light copies and Trim (not to the motion-photo strip).

A.2 Screens and wording

Document says	Code does
UI spec: onboarding "…without deleting a single photo or video" and "This version only looks…"	Body "…without deleting anything you haven't approved." and three different trust rows.
UI spec: four scan statuses incl. "Checking for hidden video…"; "Scanned 2 min ago"	Three statuses; no "scanned … ago" caption.

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Document says	Code does
UI spec: rows "JPEG photos / PNG screenshots / H.264 videos"; hero "You can free up to"; "Favorites & last 30 days — excluded"	"Camera photos / Screenshots & images / Videos"; "Shrinking can save"; no favourites row.
Design v2 §5.10: Hindi strings	Only the Settings sheet (44 strings) is translated.
Round-3 plan: five questions as a first-run card	Offered via the Get space back step, an All tools card and Settings; not at first launch.
Round-3 plan: "Running low. Make room now?"	"Running low: X GB free" / "Make room now, before the camera says it's full?"
Round-2 plan: background search indexer	Indexes only while the Search screen is open.

A.3 Clean up

Document says	Code does
Design v2 §5.5: keeper = favourite > eyes-open > sharpness > resolution > newest	That order for bursts and very-similar groups; exact groups use favourite > resolution > newest. "Newest" = date modified.
Design v2 §5.5: favourites and keep-until items excluded from lanes	The Low-quality list ignores keep-until dates after the app restarts.
Design v2 §5.5: tickets expire the day after the printed travel date	Latest printed date within −7 to +400 days of capture, expiring at the start of the next day. OTPs: keyword plus 4–8 digit number, 24 h after capture.
Design v2 §5.5: standing rules pre-select in Quick review	Expired items are always pre-selected, even with no rules on.
Round-2 plan: greetings lane for chat-app images	Greetings from any album.
Round-2 plan: photos/videos in the Download folder	Only albums named exactly "Download" or "Downloads".
Round-3 plan: chat space per folder and month	Per folder only.

A.4 Keeping it slim

Document says	Code does
Design v2 §5.6: overnight needs charging + idle	Also battery not low.
Design v2 §5.6 / round-3 plan: overnight sizes its batch to the floor gap	Gap-closing items go first; the full 300 photo / 10 video caps are still filled.
Design v2 §5.6: trips make light copies only when cooling is on	Also for albums with a Light-copy rule.
Standing-rule question 4 detail: "Overnight shrinking works toward this goal."	Overnight shrinking stays off unless switched on separately.

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A.5 Sending to a computer and privacy

Document says	Code does
Privacy page/policy: sharing "stops by itself 30 minutes after the last transfer"	It stops about 30 min after the last approved session expires (sessions expire after 10 idle minutes), so about 40 min after the last transfer; also on Stop sharing, Wi-Fi loss, app swiped away or Android's time limit.
Privacy policy: Permissions and Contact sections	Not shown in the app's Privacy page.
Receiver plan: phone shows "Verified on your Windows PC"	Phone says "…were checked on your computer and match exactly…"; the prompt names the computer's host name, not its system.
Protocol §2.5 / plan: short code shown before the fingerprint prompt	The short code appears only in the Send screen hint, not in the prompt.
Receiver plan §4: phone caches file hashes	The phone hashes the whole file every time.
Protocol §1: ports 8765–8769	Random port if all five are busy.
Protocol §7a: 413 for oversized bodies, 400 for malformed requests	The connection is closed without an answer; 400 only for a bad acknowledgement.
Phone step 3: ZIP "or a folder of your choice"	The browser's folder option is shown only in a secure context, so it is normally hidden on the plain-http page.
Design v2 §5.9: trash export covers shrink originals, clean-ups and burst extras	True, but the entry links appear only when shrink originals are in the trash.
Archive map folder "GallerySlim/*yyyy-MM*"	Matches the browser ZIP; the Receiver saves month folders directly inside the chosen folder.

A.6 GallerySlim Receiver

Document says	Code does
Plan §7: suggest the real Pictures location (or home\GallerySlim Photos if it syncs); README: the suggestion is never synced	Suggests <Pictures>/GallerySlim Photos; <home>/GallerySlim Photos if Pictures syncs; Pictures itself (with a warning) if both sync.
Protocol §7a: ids ≤ 64 characters	Receiver accepts ids up to 256 bytes; invalid or duplicate list entries are silently dropped.
README: exit 2 for bad command line or several phones	A malformed or unreachable `--phone` exits 1. Explicit `--headless` always needs `--folder`.
README/protocol: subnet probe "/24 or smaller"	/24 to /30 only; typed addresses must be IPv4 numbers.
README: 30-minute auto-quit (described under macOS)	All systems in page mode; 30–31 min; waiting for pairing counts as idle.
Plan §7: warn when a path passes 260 characters	Only a detail note on the file row, for Windows-style paths; the file is still saved and verified.
Plan: free-space check before starting	Also requires a 50 MB margin; the error's "needed" figure excludes it.
README/plan: the folder is chosen on the page	After the first choice the page cannot change it; use `--folder` or edit config.json.

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Document says	Code does
Plan §1 progress line "142 of 300 · 1.2 GB of 2.9 GB · saving to D:\Photos"	"142 of 300 photos · 1.2 GB of 2.9 GB" with "Saving to *folder*" on its own line.
App message "N of M files could not be received. Press Try again to retry them."	Not shown on the Done view; the retry is "Receive more".

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Appendix B Behaviour to confirm on a device

These behaviours follow from the code but look unintended, are inconsistent, or could not be confirmed without a device. They are good starting points for test cases.

#	Area	Observation
1	First launch	After granting access in Android settings, the onboarding screen may stay until the app is reopened.
2	Progress	Swiping the app away during a manual batch may cancel it.
3	Notifications	The trip result uses the same slot as manual-batch progress and may be replaced.
4	Ready to clear	Items whose originals were moved to the trash via "Send to laptop" stay listed; Keep original then says the original is already in the trash.
5	Quick review	After a successful trash the summary says "You kept everything."; "Review again" still contains the trashed items.
6	Clean up	"Keep selected for" on Low-quality photos is forgotten after an app restart.
7	Get space back	"Try one first" skips the Google Photos warning.
8	Trash export	No entry point when the trash holds only clean-up items.
9	Your month	The notification is about last month but opens the current month.
10	Wording	"Send to laptop" vs "Send to computer"; "Scan the laptop's code" vs "Scan the computer's code"; Balanced "(recommended)" but Gentle is the default.
11	Plurals and numbers	"Trip prep: 1 items shrunk", "Full in about 1 days at this rate", "1 photos" in Search; Make room's "Shrink N items" has no thousands separator.
12	Units	Widget, low-space alert and monthly card use decimal GB; app screens use binary GB.
13	Browser page	"Saved as one file ✓" and "Check your Downloads folder" appear after 1.5 s, before the download finishes.
14	Receiver (gs2)	The phone's progress bar may never reach "All … sent" when the Receiver verifies files it already had.
15	Receiver page	"Saved, checking…" is also the final label for unconfirmed files; the "phone ended the session" note is never displayed.
16	Home	"Your gallery is already in good shape" is followed by "without losing a single photo."
17	Media library naming	The exact de-duplicated names of lossless JPEG copies, stripped motion photos and .mp4 copies (e.g. "name (1).mp4") depend on Android.

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Glossary

Album rule
A per-album setting: Default, "Light copy for photos older than N years" (1/2/3/5) or "Never touch".

All tools
The full dashboard behind the minimal Home screen.

Android's trash
The system's recycle bin for photos and videos. Items stay restorable for 30 days, then Android deletes them. Space is freed only when they leave the trash.

App lock
Optional fingerprint/PIN prompt before Ready to clear and the vault.

Banding guard
A video check that rejects copies showing visible colour steps ("banding") that the original does not have.

Burst
At least 3 photos in the same album taken at most 1.5 s apart.

Clean up list (lane)
One of the suggestion lists: Old screenshots, Old chat media, Low-quality photos, Expired, Old receipts & tickets, Greetings & forwards, Old downloads.

Contact sheet
One grid image showing every shot of a burst, saved before the extra shots go to the trash.

Cooling
The setting "Lighter copies of old photos": photos older than 1–3 years in Default albums become 2048 px light copies when shrunk.

D1, D2
The only two permanent-delete paths: D1 "Free this space now", D2 "Send the trash to computer" (Receiver-verified files only).

Fully sent
Browser (gs1) only: every byte of the file was written to the network. It does not prove the computer saved it.

gs1
The browser way of sending to a computer: a page served by the phone, a ZIP or single downloads; unencrypted on the local network.

gs2
The GallerySlim Receiver protocol: QR pairing, signed requests, encrypted transfer, on-disk verification.

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HEIC / HEVC
Efficient photo (HEIC) and video (HEVC, H.265) formats that Android 12+ shows natively. GallerySlim's copies use them.

ID shield
On-device text reading that keeps identity documents and unreadable photos out of the Clean up lists.

Keep original
Rejects a smaller copy: GallerySlim deletes its own copy and leaves the original exactly as it was.

Keep until
A date after which an item appears in the Expired list; set by "Keep selected for" or by the expiry reader.

Keeper
The copy in a duplicate group that will be kept; chosen automatically, changeable by long-press.

Ladder of regret
The order of actions from most to least reversible: copy → Ready to clear → trash → delete for good.

Ledger
The app's count of confirmed shrink originals that are freed vs still in the trash.

Light copy
A preset (and Big videos action) that also reduces size: photos to 2048 px on the long edge, videos to 720 px on the short edge.

Lossless re-optimisation
Re-packing a JPEG's data more efficiently without changing a single pixel; used for Ultra HDR and as a fallback.

Make room
A plan to free a target amount, gentlest steps first.

Motion photo
A photo file with a short video clip hidden inside (Samsung, Pixel).

Never touch
An album rule that excludes the album from every shrink, light copy and trim.

Original
The photo or video as it was before GallerySlim made a smaller copy.

Preset
Gentle, Balanced, Aggressive or Light copy: the minimum quality a copy must reach.

Protect window
"Protect new photos for" 7/30/90 days: newer photos and favourites start unticked.

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Quick review
One swipe queue over all suggestions and duplicate extras, ending in one trash dialog.

Ready to clear
The list of originals whose smaller copies are done and checked, waiting for the user to clear them to the trash.

Receiver
The GallerySlim Receiver laptop program (Windows, macOS, Linux).

Run
For Undo this run: the set of originals cleared in one Ready-to-clear dialog. For delete for good: one sending session since the queue was set.

Short code
Two letters and two digits (e.g. "K7-42") shown on the phone and by the Receiver, so the user can tell phones apart. It is not a password.

Smaller copy
The new file GallerySlim makes; labelled "GallerySlim" in Compare.

Snackbar
A short message at the bottom of the app screen.

Spot check
The video quality check: 5 frames, each must score at least 0.90.

SSIM, MS-SSIM, SSIMULACRA2
Measures of how similar two images look. GallerySlim requires every photo copy to reach the preset's floor on these.

Standing rules
The answers to the five questions: they pre-select suggestions, set the storage goal and cooling, and enable the weekly nudge.

Storage floor
"Keep this much space free": a goal that overnight shrinking and Make room work toward.

Trip mode
"Make room by a date": a one-off shrink the evening before a chosen date.

Ultra HDR
A JPEG with an extra HDR "gain map" layer (recent Pixel and Samsung phones). GallerySlim never re-encodes it.

Vault
Optional exact copies of originals in a user-chosen folder, made before shrinking.

Verified
gs2 only: the Receiver re-read its saved file from disk, and its SHA-256 and size equal the phone's whole current file.

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